Implementing MLFFs/MLIPs on Molten Cu-In Nanoparticle Systems to Model Growth of Carbon Nanotubes



Supervised by:
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Department of Chemical Engineering
IIT Kanpur

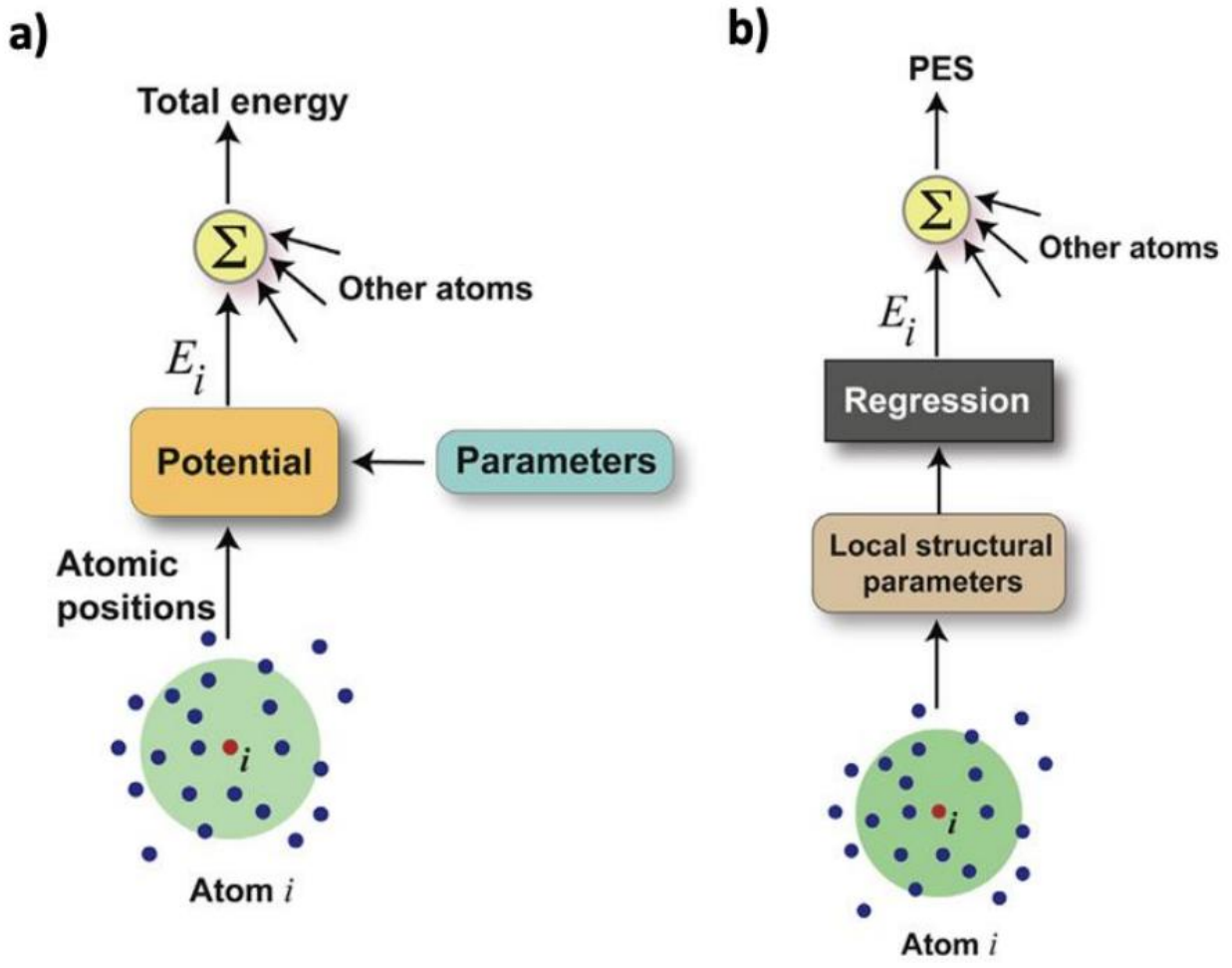
Anmol Bansal
Sourish Das
Shreshth Gupta

Part 1:
Introduction to Machine-Learned Force Fields (MLFFs) / Interatomic Potentials (MLIPs)

What are MLFFs/MLIPs?

Introduction

- Atomic configurations → Potential Energy Surface
- Input: Coordinates/vectors + element types
  Output: Spatial derivatives of PES
- Atomistic Simulations:
  - AIMD
  - PBP
- MLIPs are the Bridge!

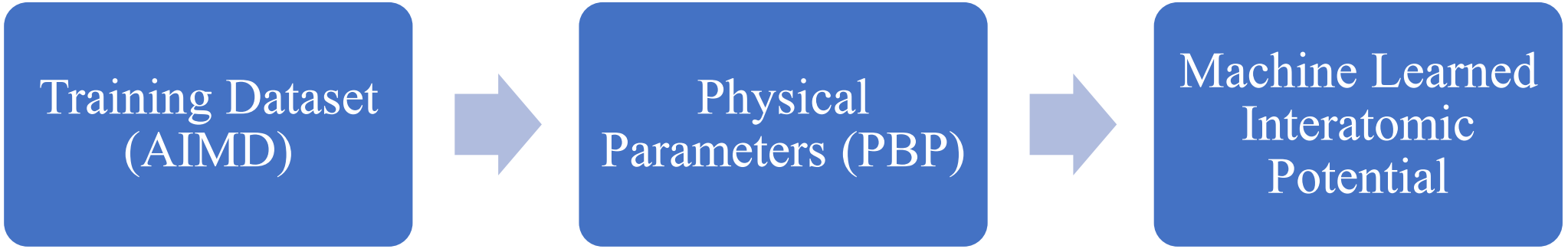


What are MLFFs/MLIPs?

Motivation and Need

- Computational infeasibility of AIMD in large chemical systems and longer timescales.
- Less accuracy of PBP-based MD for materials.
- MLIPs can be supplemented with physics functionals for extrapolating complicated energy profiles.

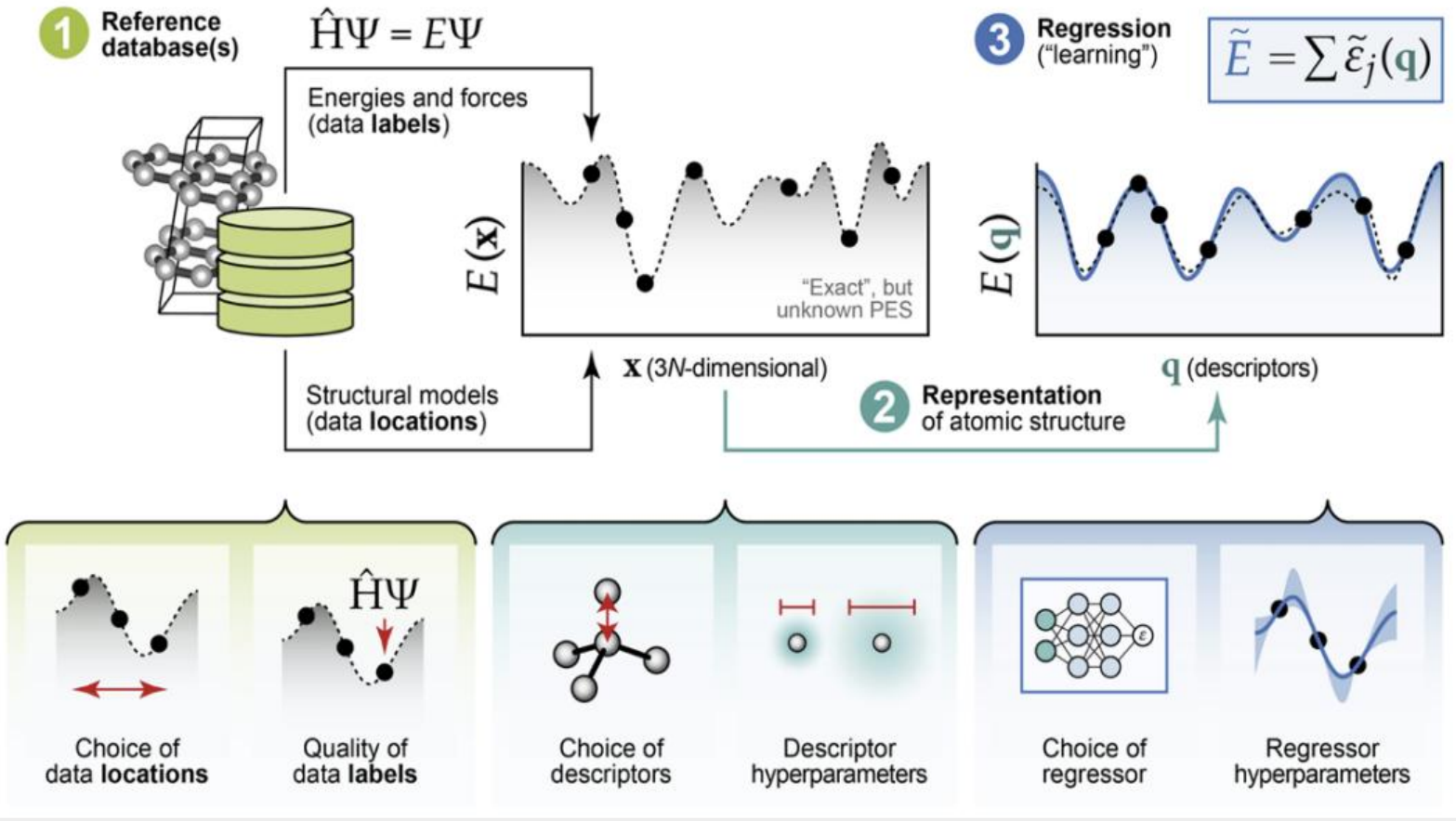
- MLIPs: Best of Both Worlds.



What are MLFFs/MLIPs?

Basic Principles

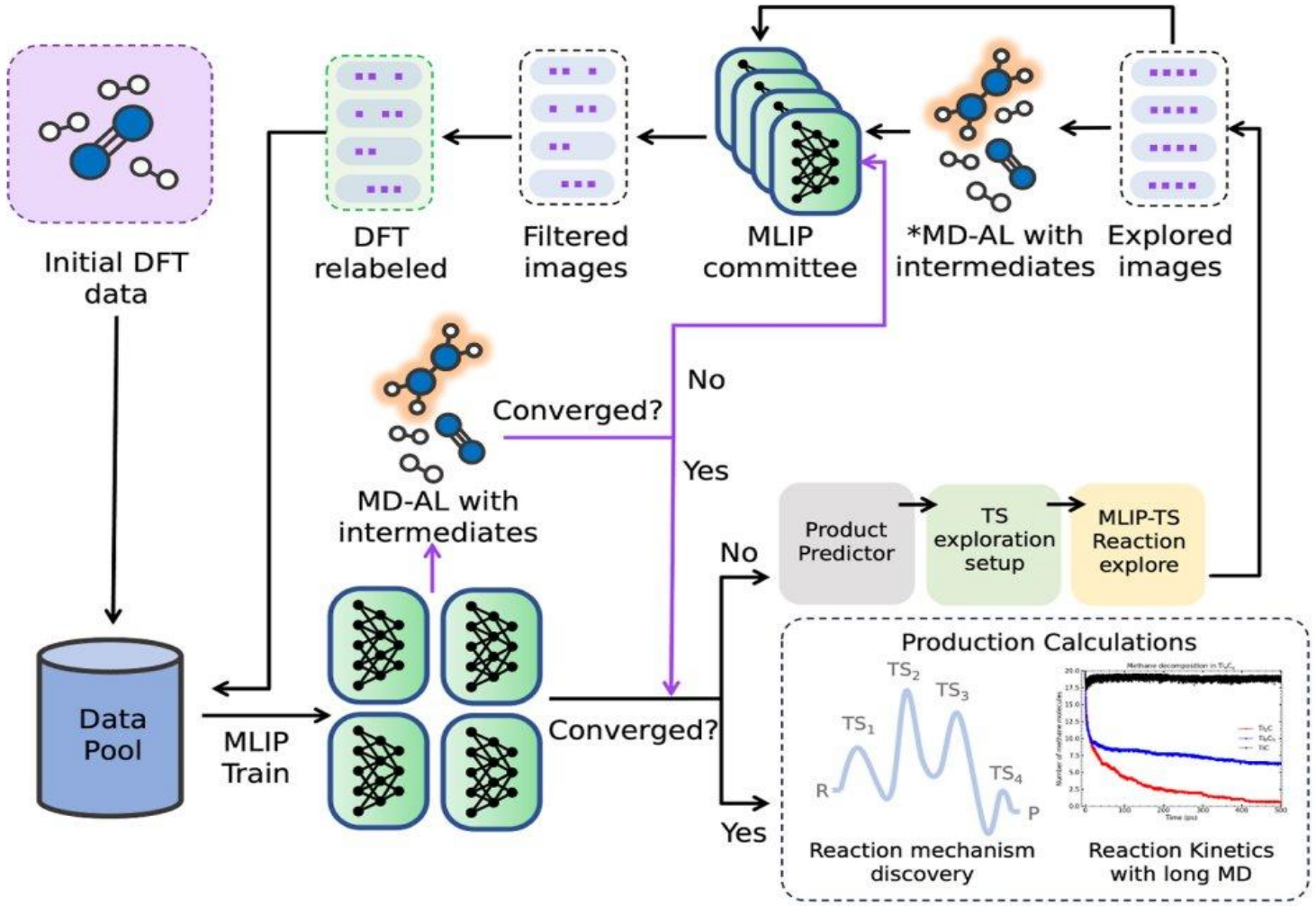
- Atomic Environment Featurization
- Descriptors
- Symmetrical Invariance
- Unification of Explicit and Implicit AEF



Literature on MLFFs – Molten Systems, Nanomaterials, CNT Growth, etc.

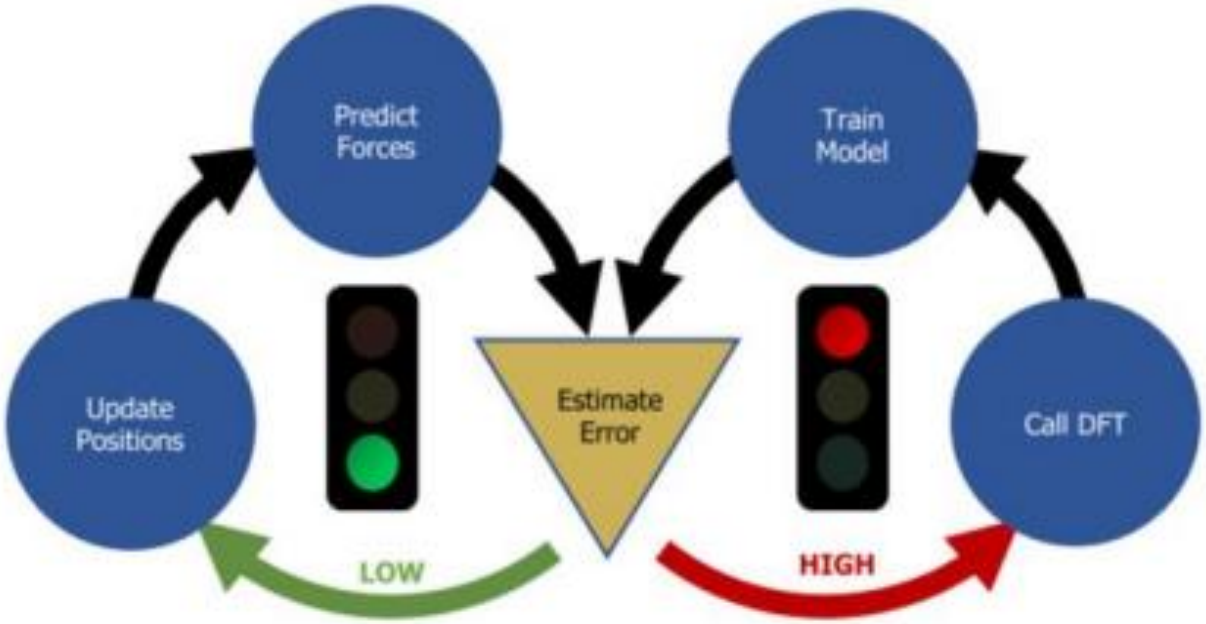
Reactive Active Learning

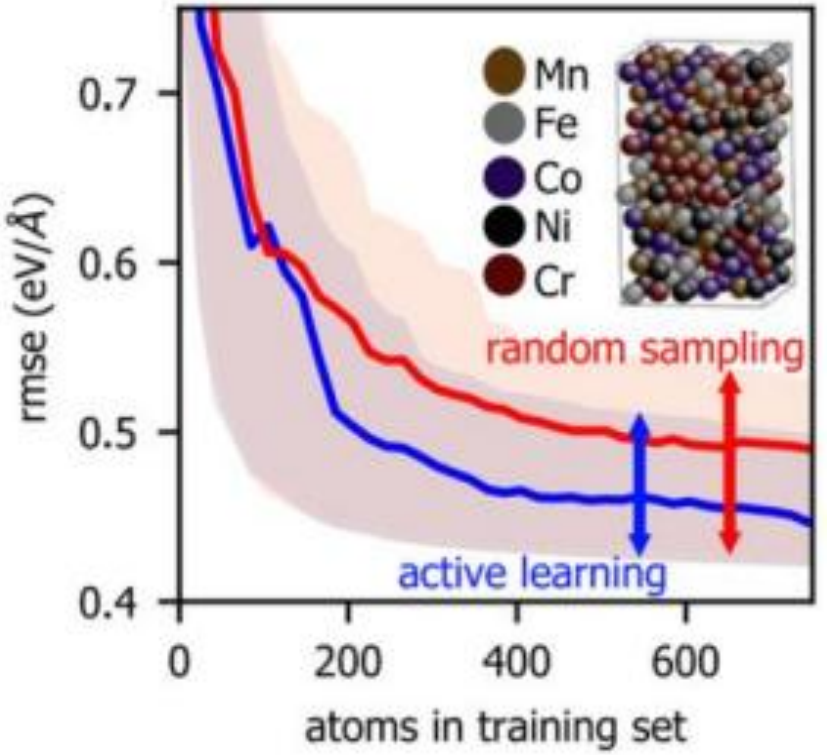
- Traditional MLIP training paradigms ill-equipped to handle rare events.
- *Workflow*:
  - Automated Reaction Exploration
  - Uncertainty Driven Active Learning
  - Transition State Sampling
- Benchmarking:
  - Gas Phase
  - Solution Phase
  - Heterogeneous



Literature on MLFFs – Molten Systems, Nanomaterials, CNT Growth, etc.

Reactive Active Learning





Literature on MLFFs – Molten Systems, Nanomaterials, CNT Growth, etc.

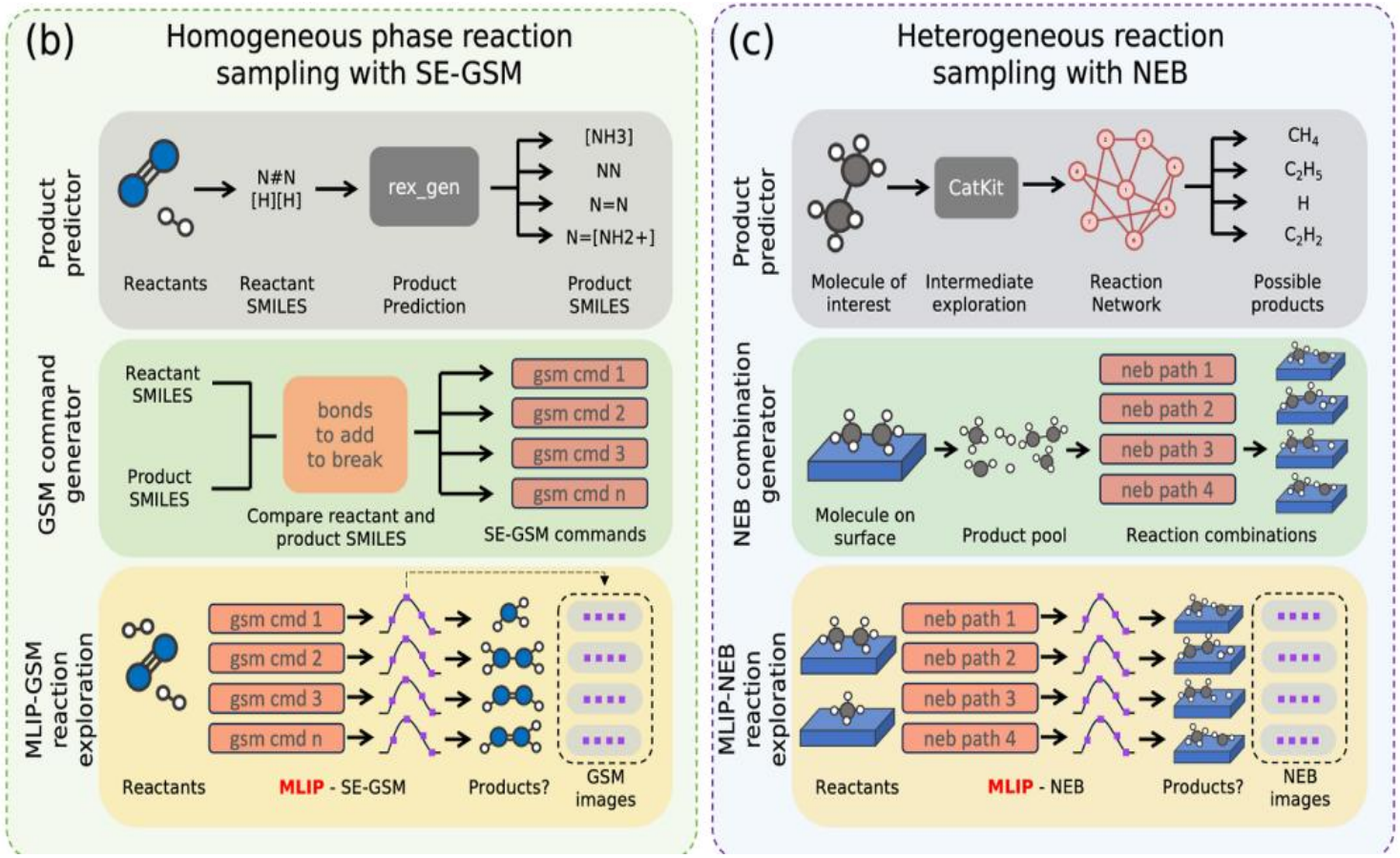
Reactive Active Learning

- *Methodology:*

  - Initial molecules for bootstrap dataset.
  - Short AIMD
  - MLIP training
  - Configuration filtering

- Committee of MLIPs:

  - Product prediction from reactant SMILES.
  - TS/Intermediates setup through SE-GSM and/or NEB.
  - MLIP driven reaction exploration.



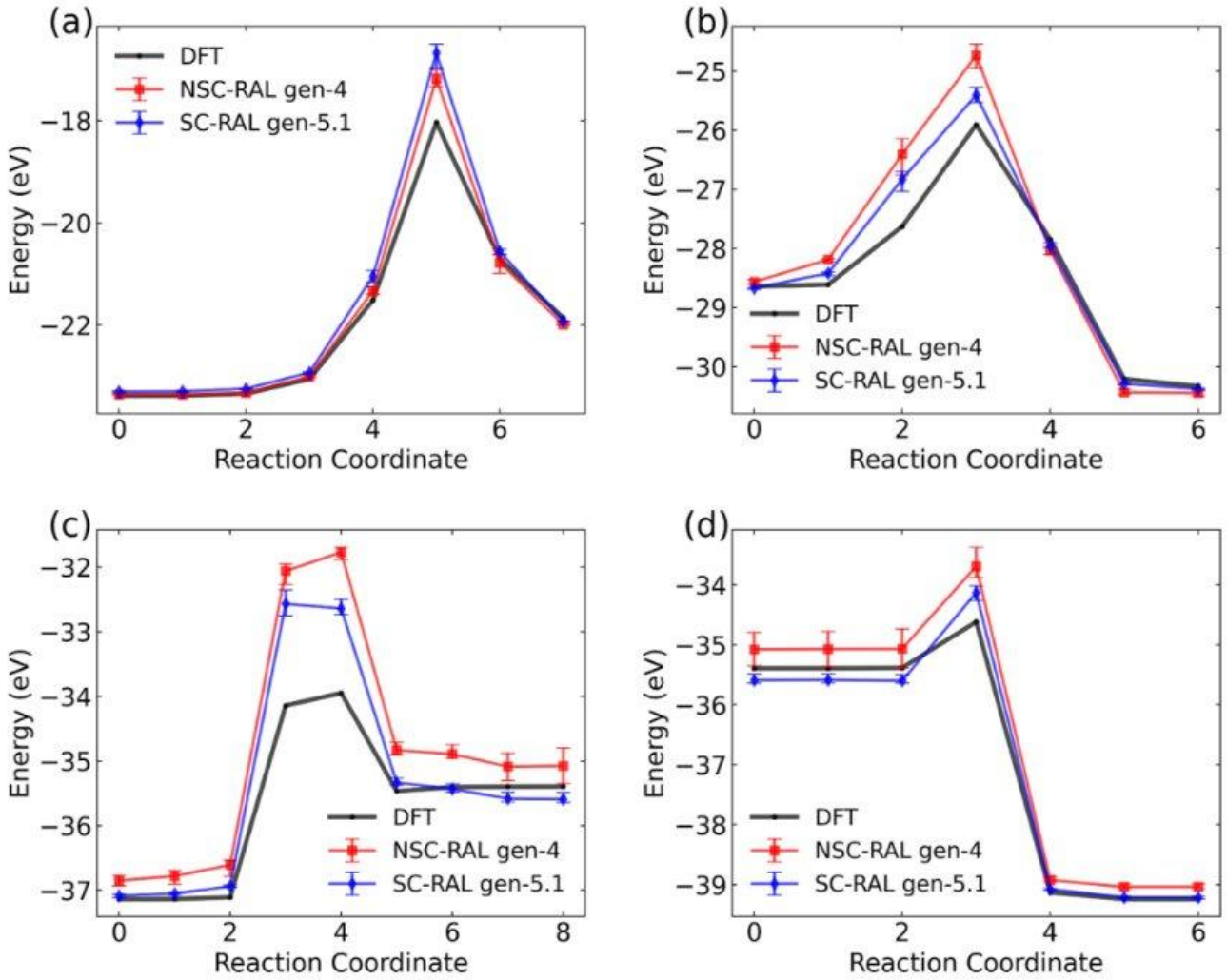
Literature on MLFFs – Molten Systems, Nanomaterials, CNT Growth, etc.

Reactive Active Learning

- Uncatalyzed Gas-Phase Ammonia Synthesis

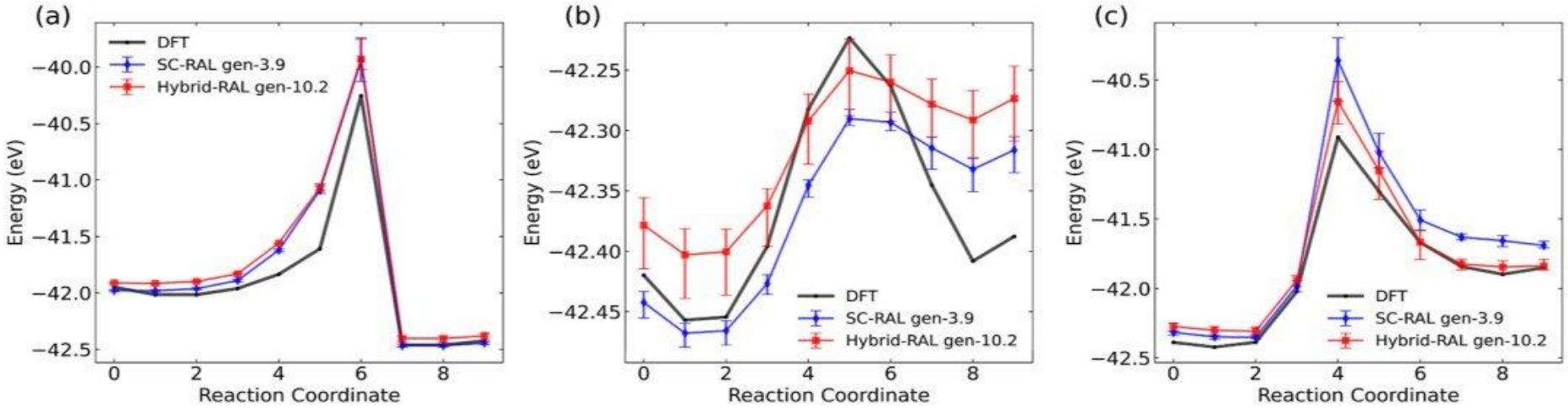
- Reactions:
  - $N_2 + H_2 \rightarrow cis- HNNH$
  - $cis- HNNH + H_2 \rightarrow H_2NNH_2$
  - $H_2NNH_2 + H_2 \rightarrow HNNH_3 + H_2$
  - $HNNH_3 + H_2 \rightarrow NH_3 + NH_3$.

- Both NSC-RAL and SC-RAL MLIPs show significant errors in predicting the forward and reverse barriers, indicating poor performance in estimating the transition state energy.



Literature on MLFFs – Molten Systems, Nanomaterials, CNT Growth, etc.

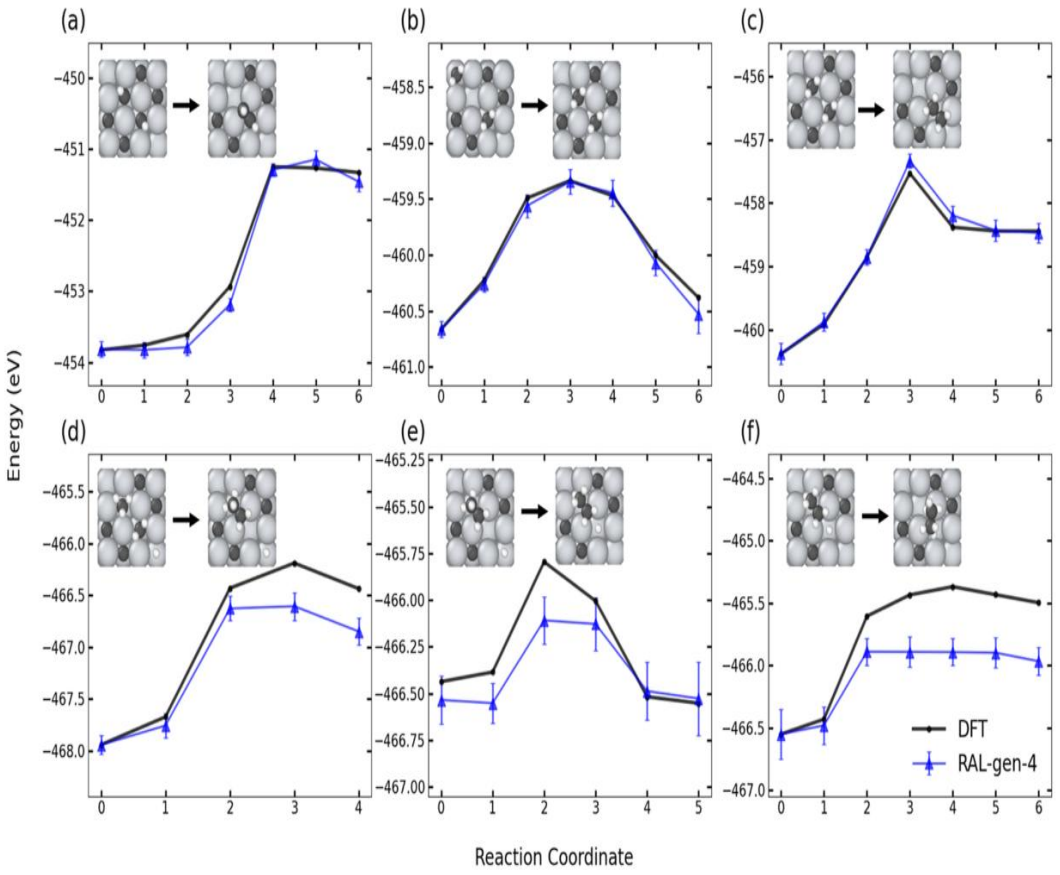
Reactive Active Learning



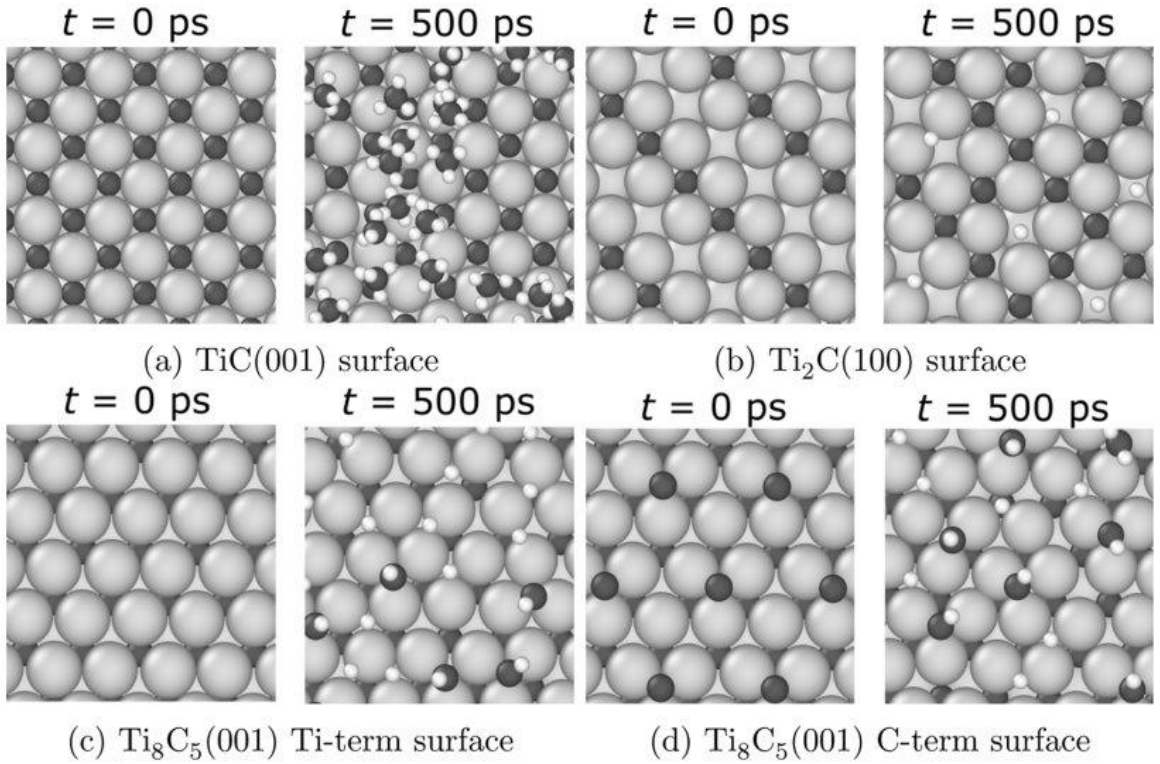
- Hydrolysis of Methanamine
- Reactions:
  - $CH_2NH + H_2O \rightarrow NH_2CH_2OH$
  - $NH_2CH_2OH \rightarrow NH_2CH_2OH$
  - $NH_2CH_2OH \rightarrow CH_2O + NH_3$
- Both MLIPs predict similar reaction PES as well as forward and reverse barriers. Although the absolute error is small for this reaction, the reaction barrier itself is very low, which may result in large a percentage error.

Literature on MLFFs – Molten Systems, Nanomaterials, CNT Growth, etc.

Reactive Active Learning



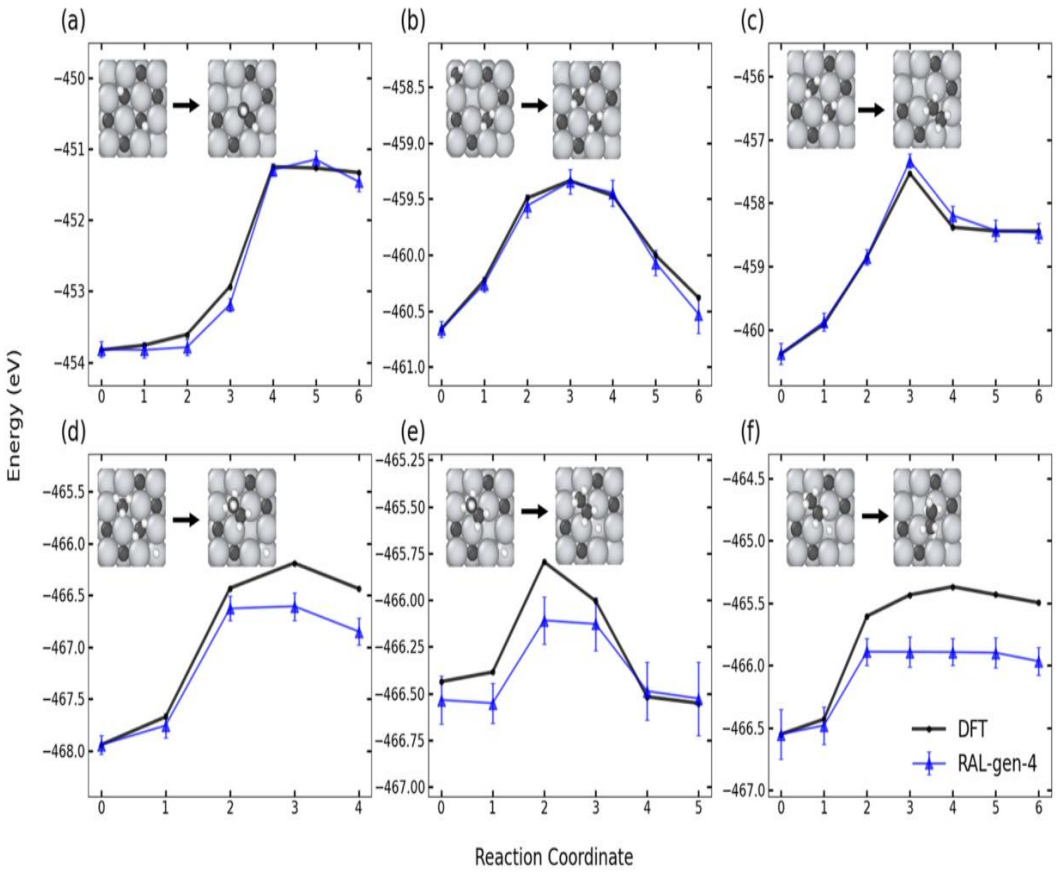
Methane Activation and C−C Coupling Surface Catalysis Reaction on Titanium Carbide



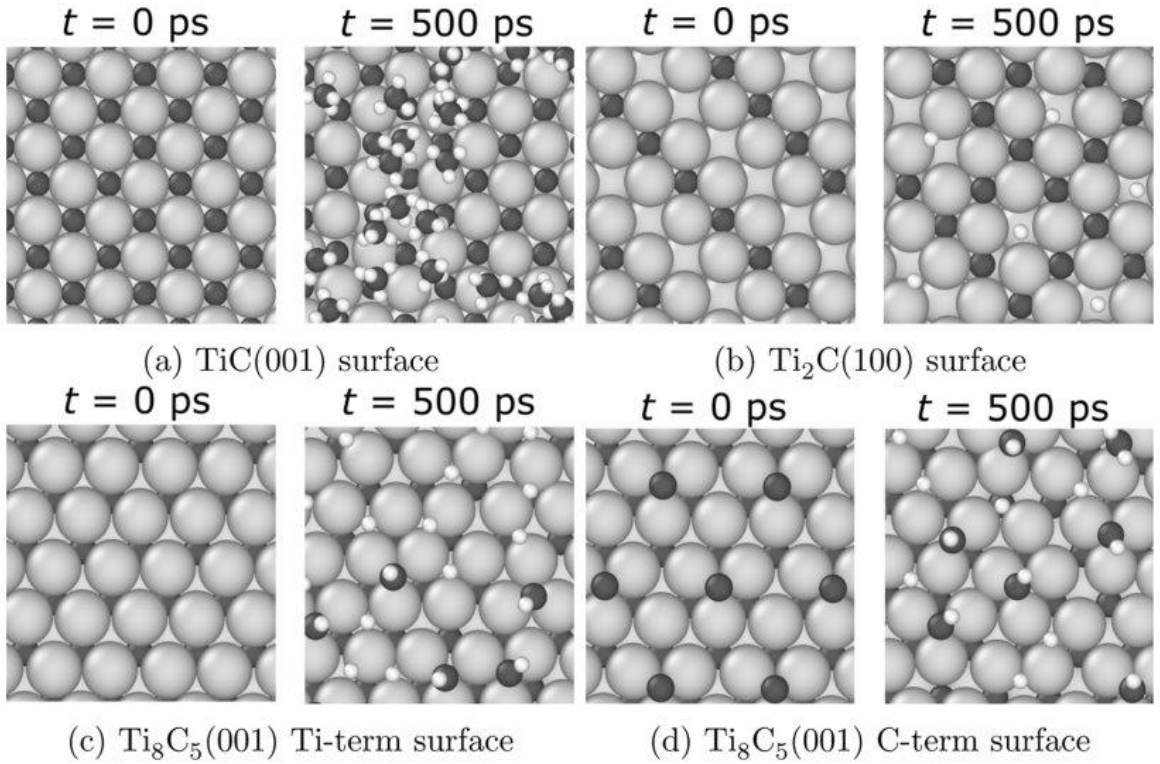
Snapshots of TixCy surfaces before (t = 0 ps) and after (t = 500 ps) T = 1200 K NVT-MD simulations initialized with 20 CH4 molecules. Ti atoms shown in light gray, C in dark gray, H in white.

Literature on MLFFs – Molten Systems, Nanomaterials, CNT Growth, etc.

Reactive Active Learning



Methane Activation and C−C Coupling Surface Catalysis Reaction on Titanium Carbide

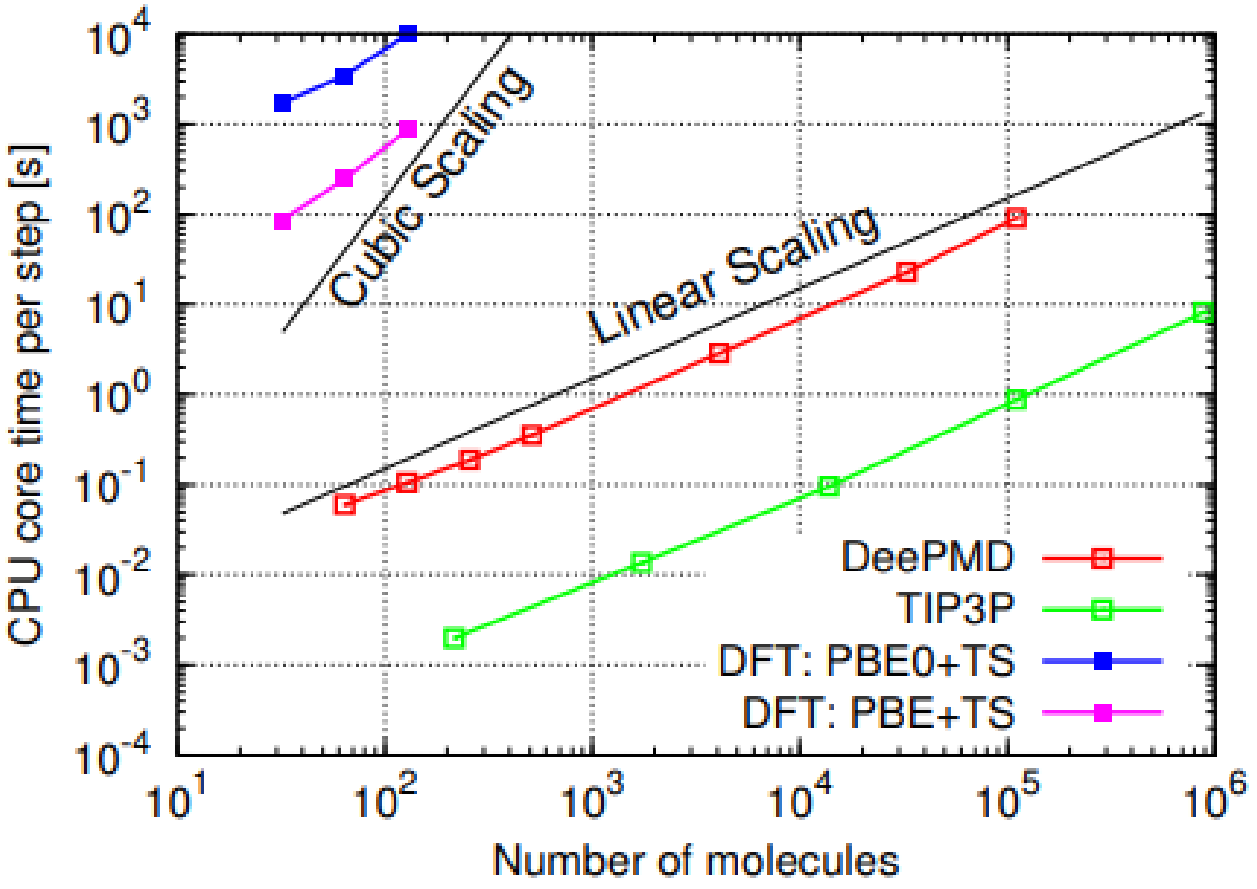


Snapshots of TixCy surfaces before (t = 0 ps) and after (t = 500 ps) T = 1200 K NVT-MD simulations initialized with 20 CH4 molecules. Ti atoms shown in light gray, C in dark gray, H in white.

Implementation Toolkit

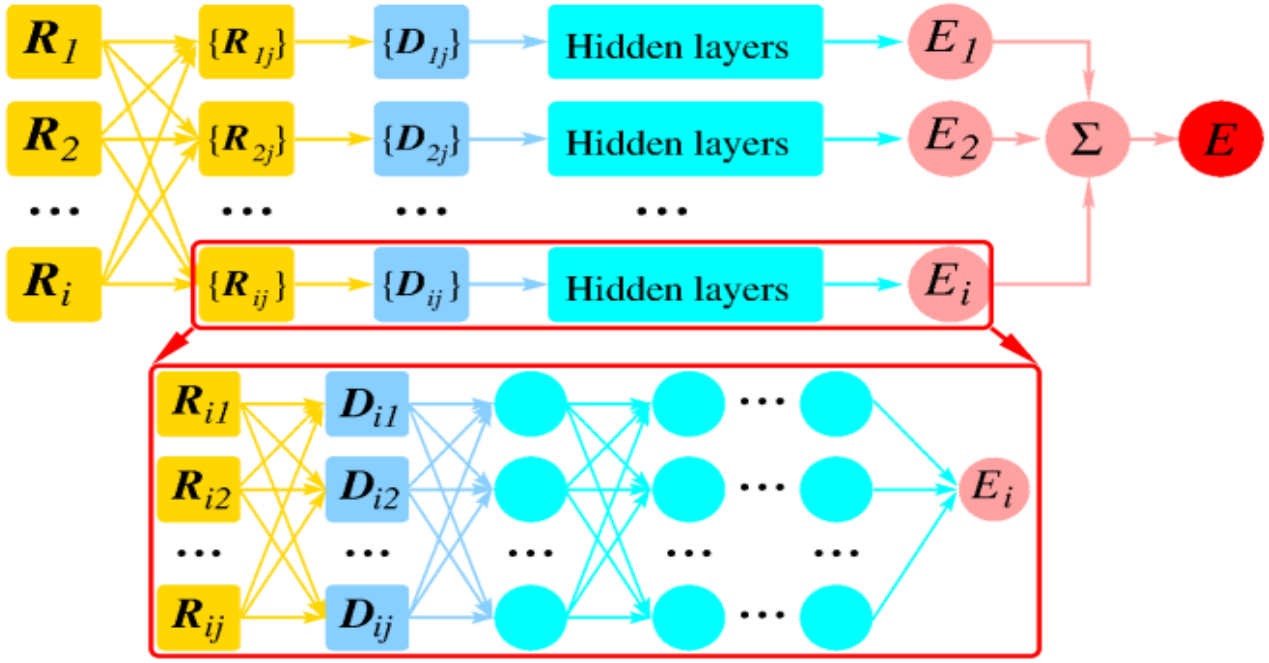
DeePMD-kit

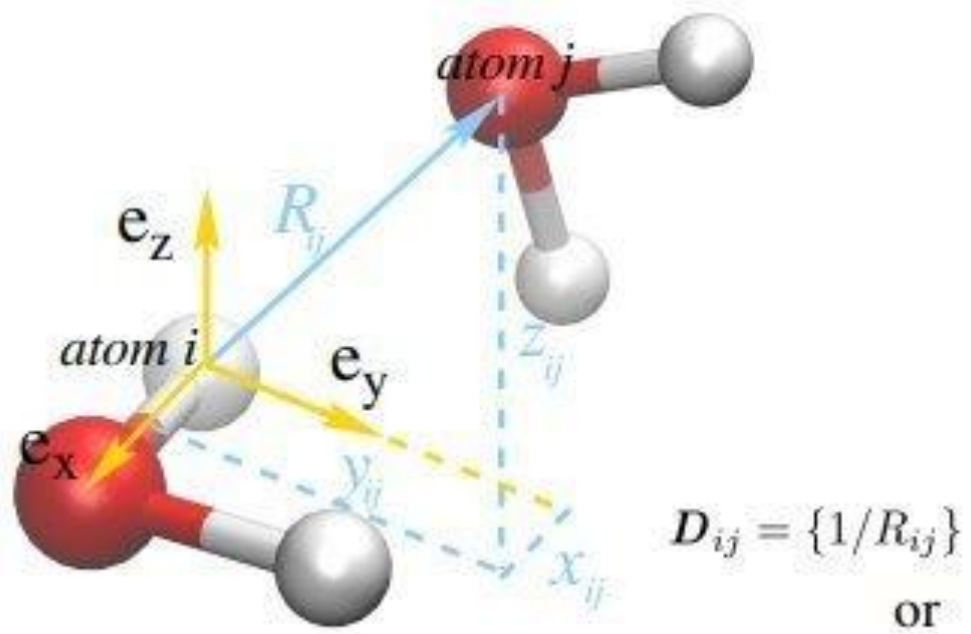
- Deep neural networks for MD.
- Local reference frame and local environment assigned to each atom.
- Bypassing auxiliary quantities for preserving symmetries.
- Flexible family of loss functions.
- Computational cost scales linearly with the number of atoms.
- Can be parallelized due to local decomposition and the near-neighbor dependence of its “atomic energies”.



Implementation Toolkit

DeePMD-kit



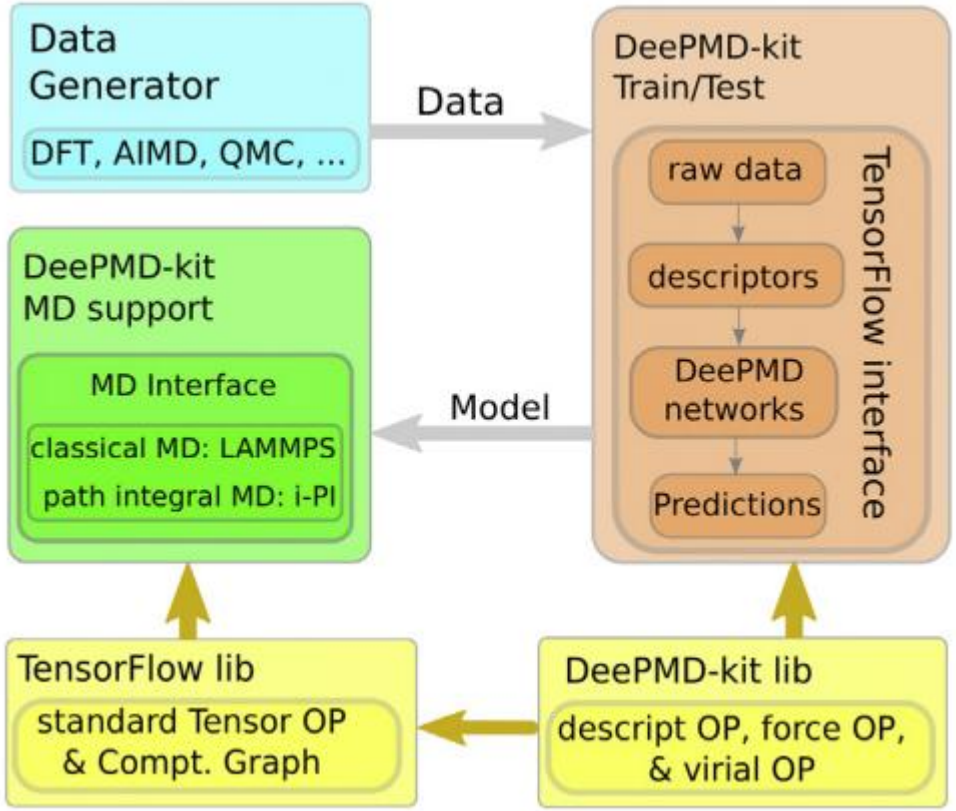


- Local coordinate frame (***Dij***) is set up for every atom a neighbors inside ***Rc***. This preserves translational, rotational, and permutational symmetries. ***Dij*** contains radial and angular information of neighbouring atom.

- ***Dij*** is the FF-DNN input. A linear transformation is applied to the input data, followed by activation with hyperbolic tangent function. There are three hidden layers in our network. We use SGD with the Adam optimizer.

Implementation Toolkit

DeePMD-kit

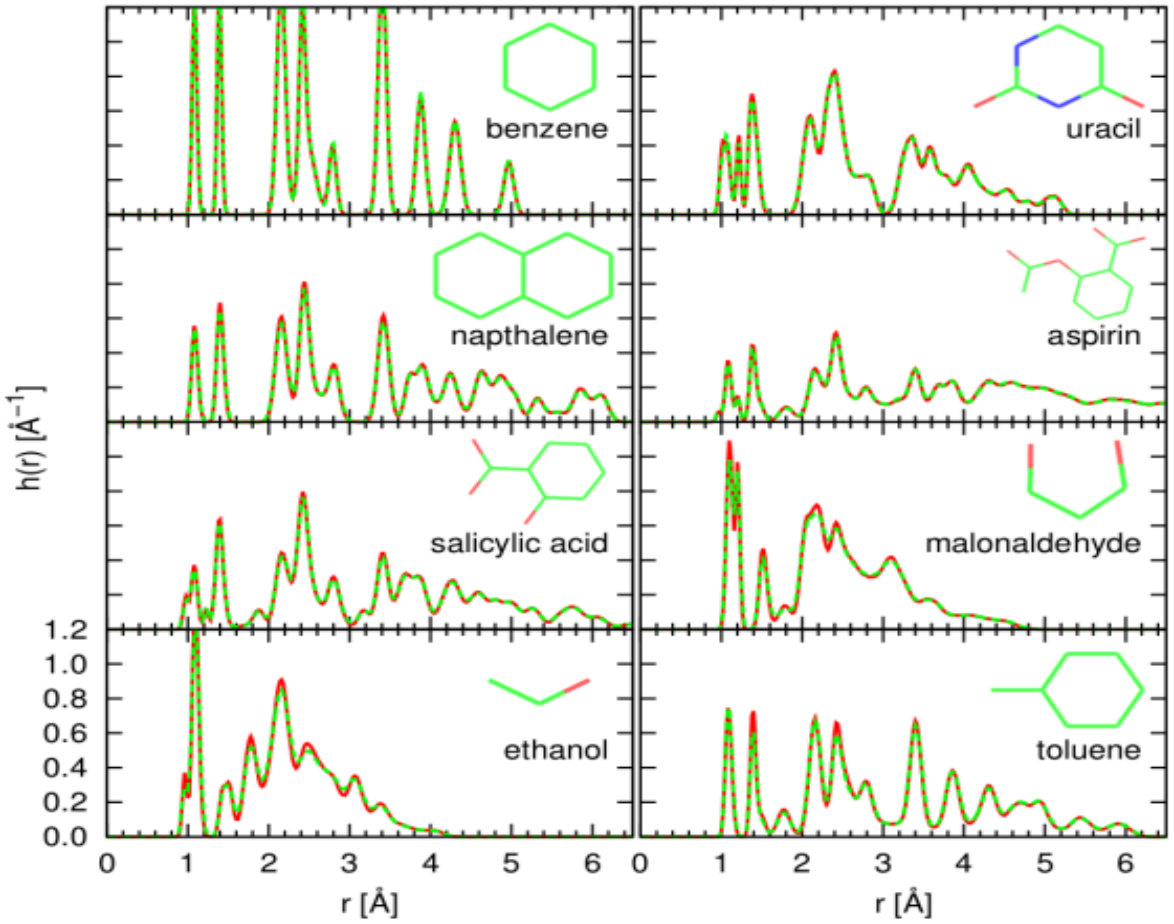


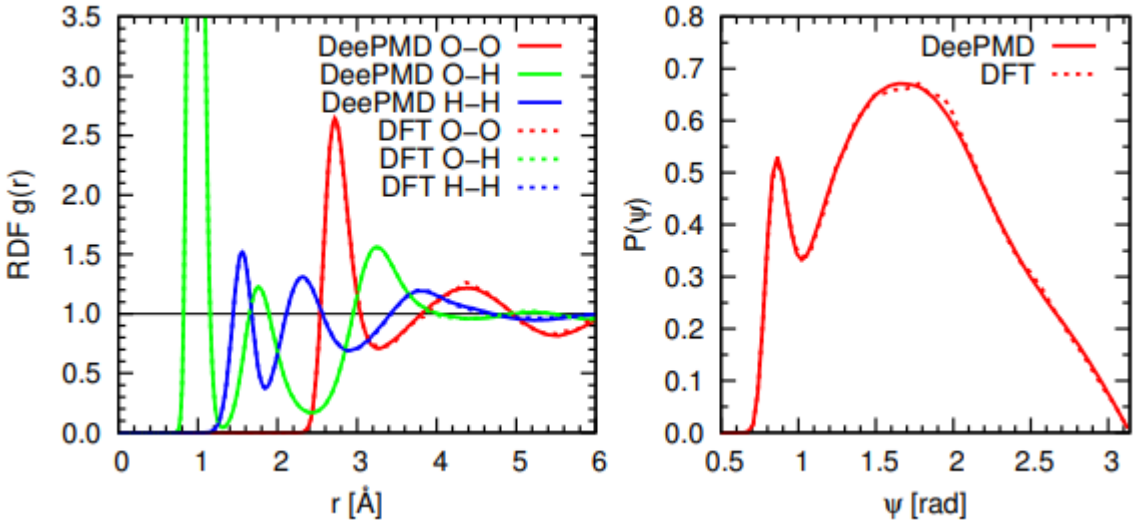
Implementation Toolkit

DeePMD-kit

Implementation Toolkit

DeePMD-kit





Part 3:

Method

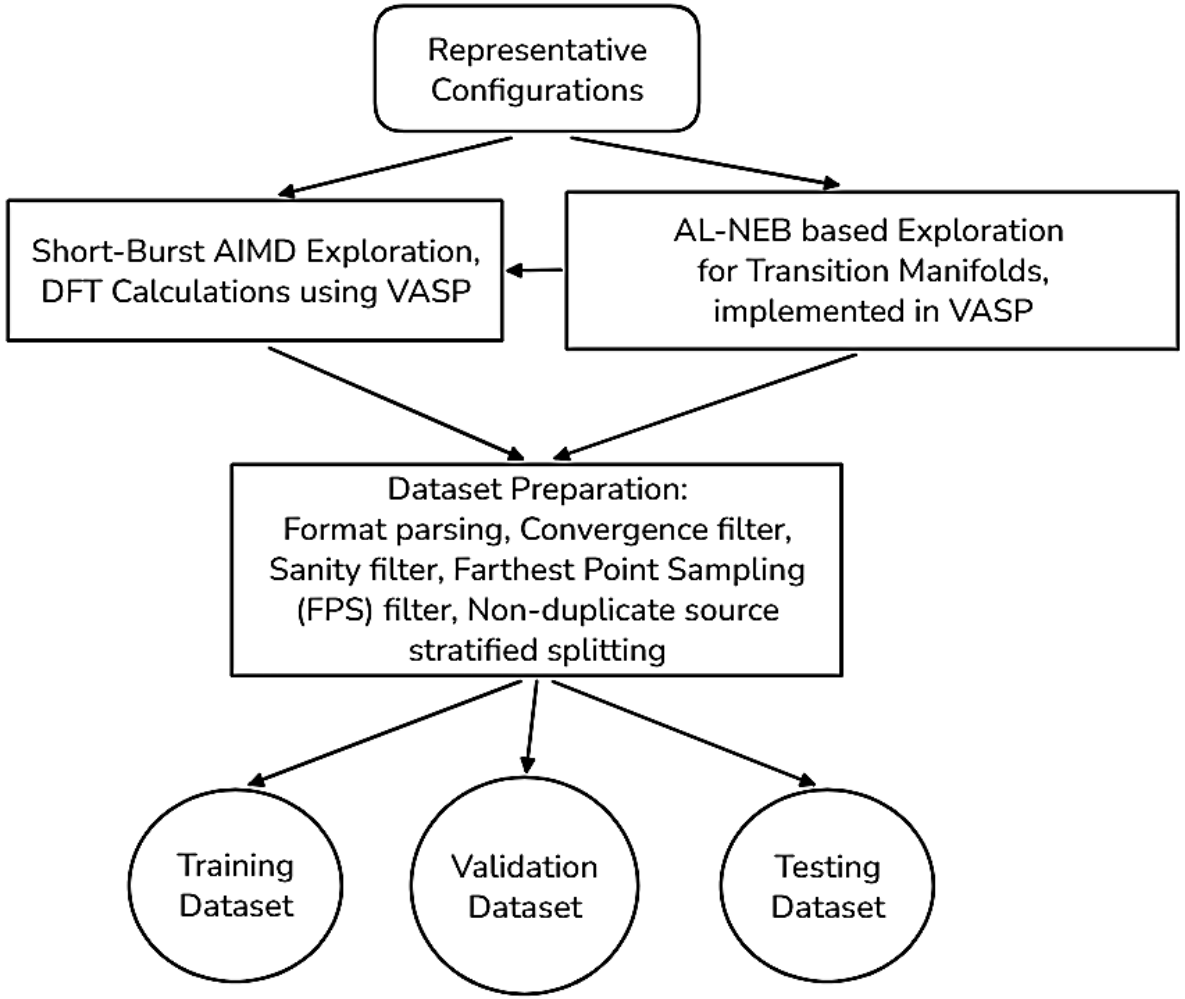
Developing MLFF for a specific chemical system

Types of Data



Anmol &
Sourish

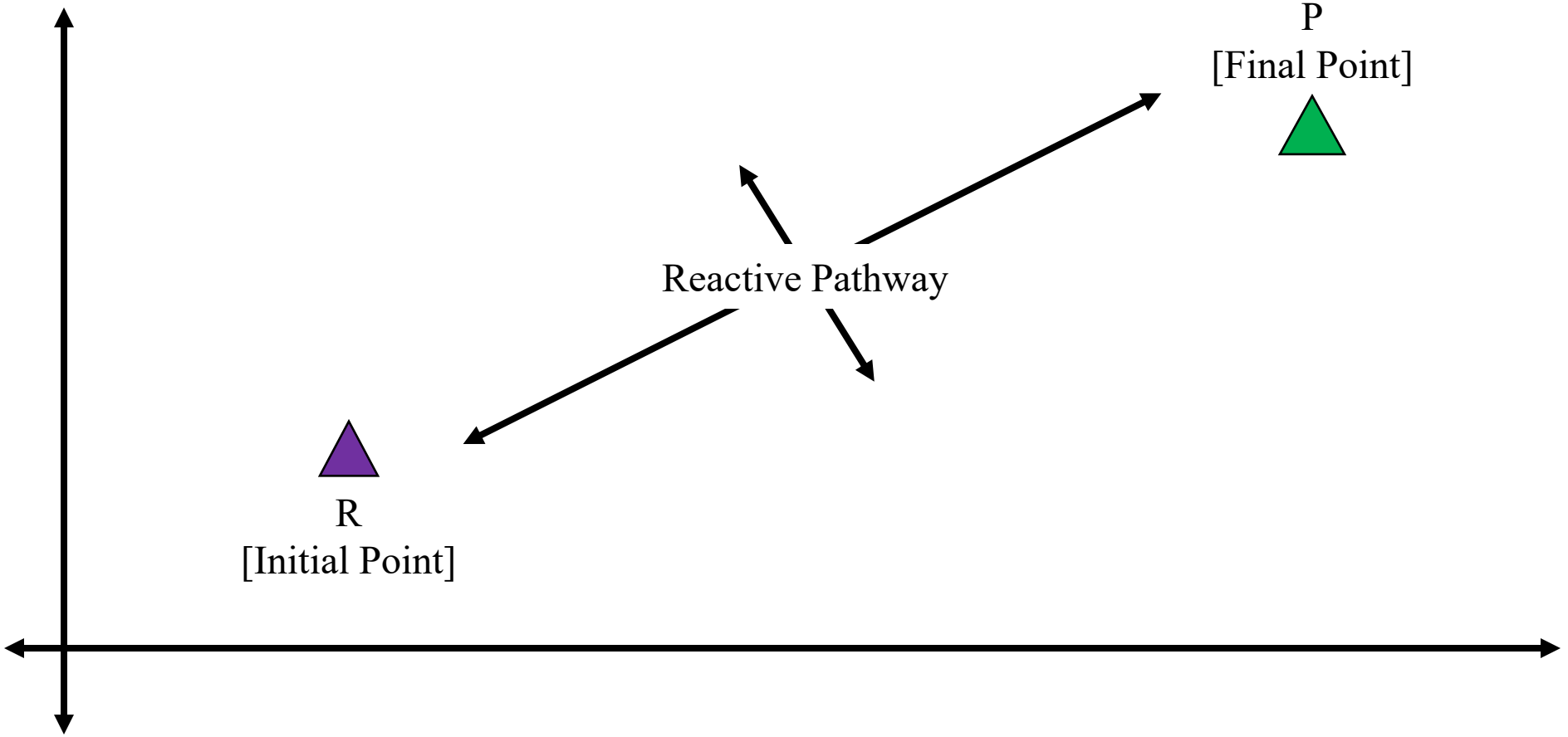
Data Generation



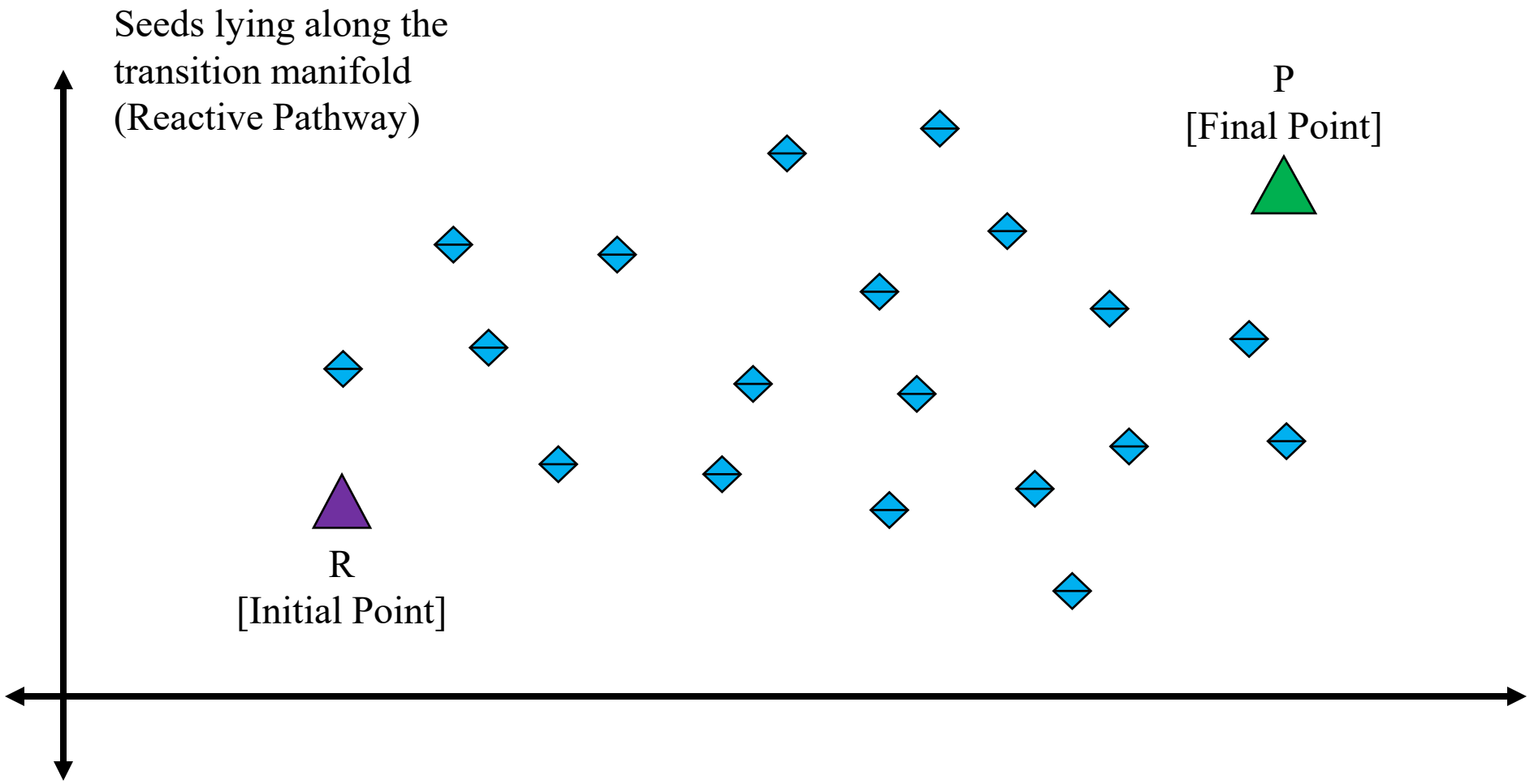
Anmol

Idea of an "**ALNEB-guided Swarm AIMD**"

- Using exploration-exploitation phase of AL-NEB.
- Collecting data points at regions of most uncertainty.
- No need for renunciation phase.
- Avoiding seeds from unnecessary regions.



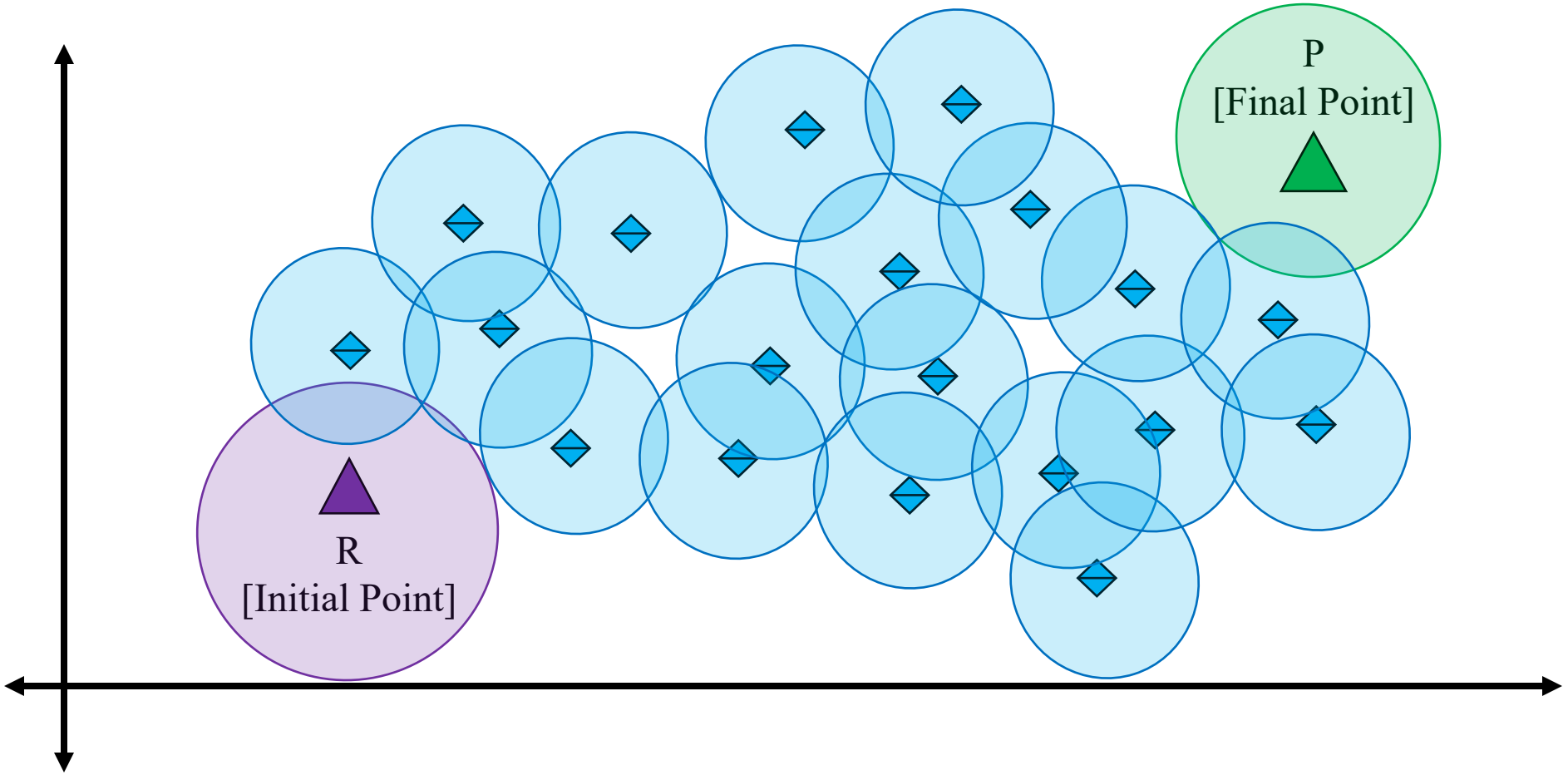
Idea of an "**ALNEB-guided Swarm AIMD**"



Idea of an "**ALNEB-guided Swarm AIMD**"

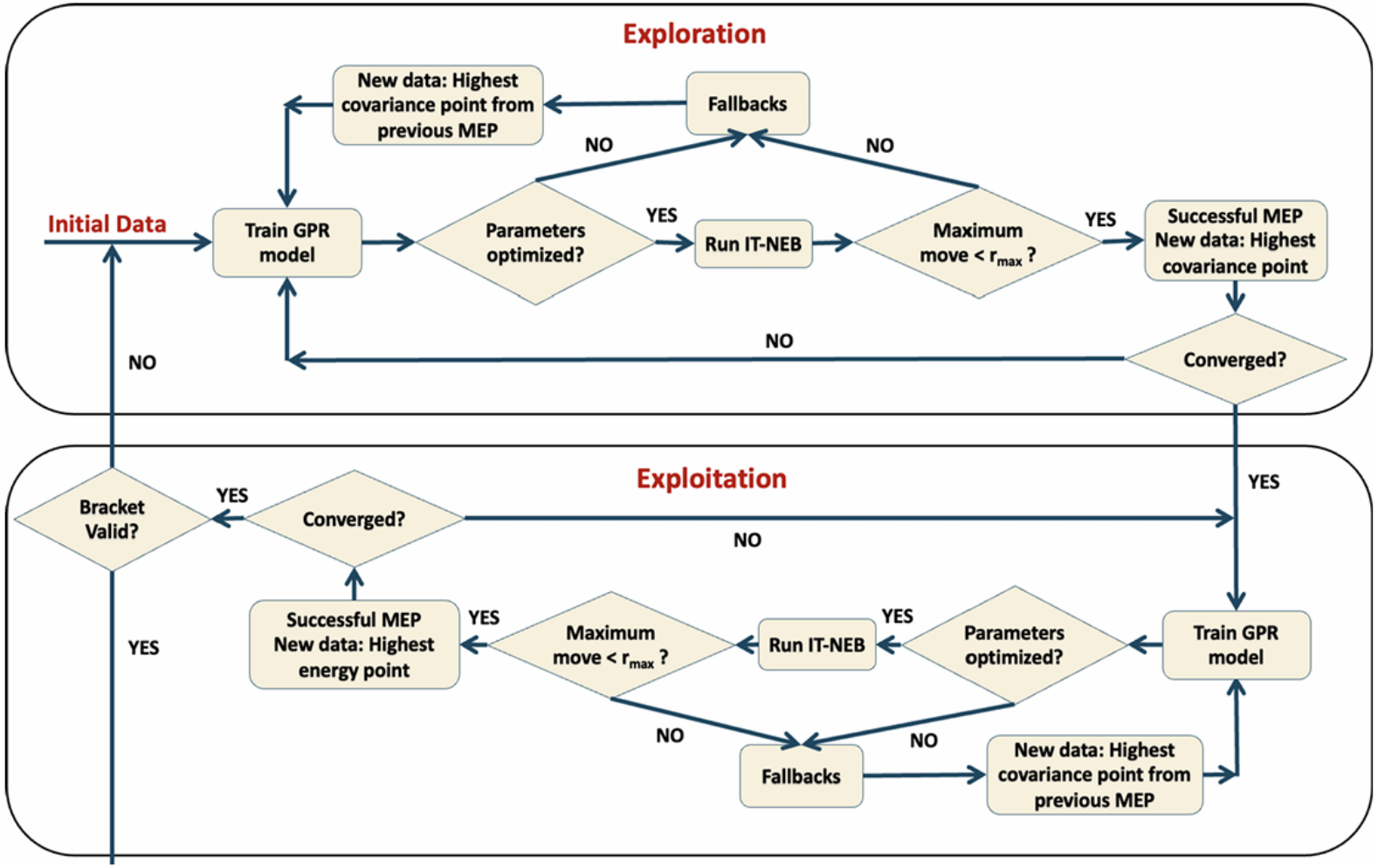
- Short AIMD trajectories running with different initial velocity states covering the configurations around the seeds like a swarm.

- All seeds from the ALNEB run are used and data from the transition manifold is covered.

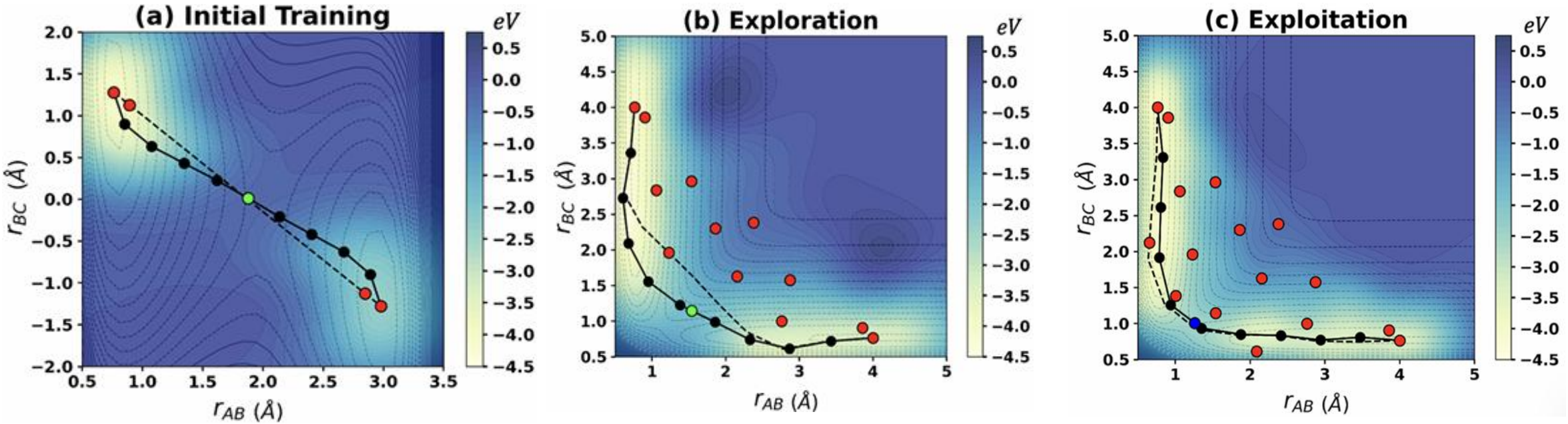


How does ALNEB work in our favour?

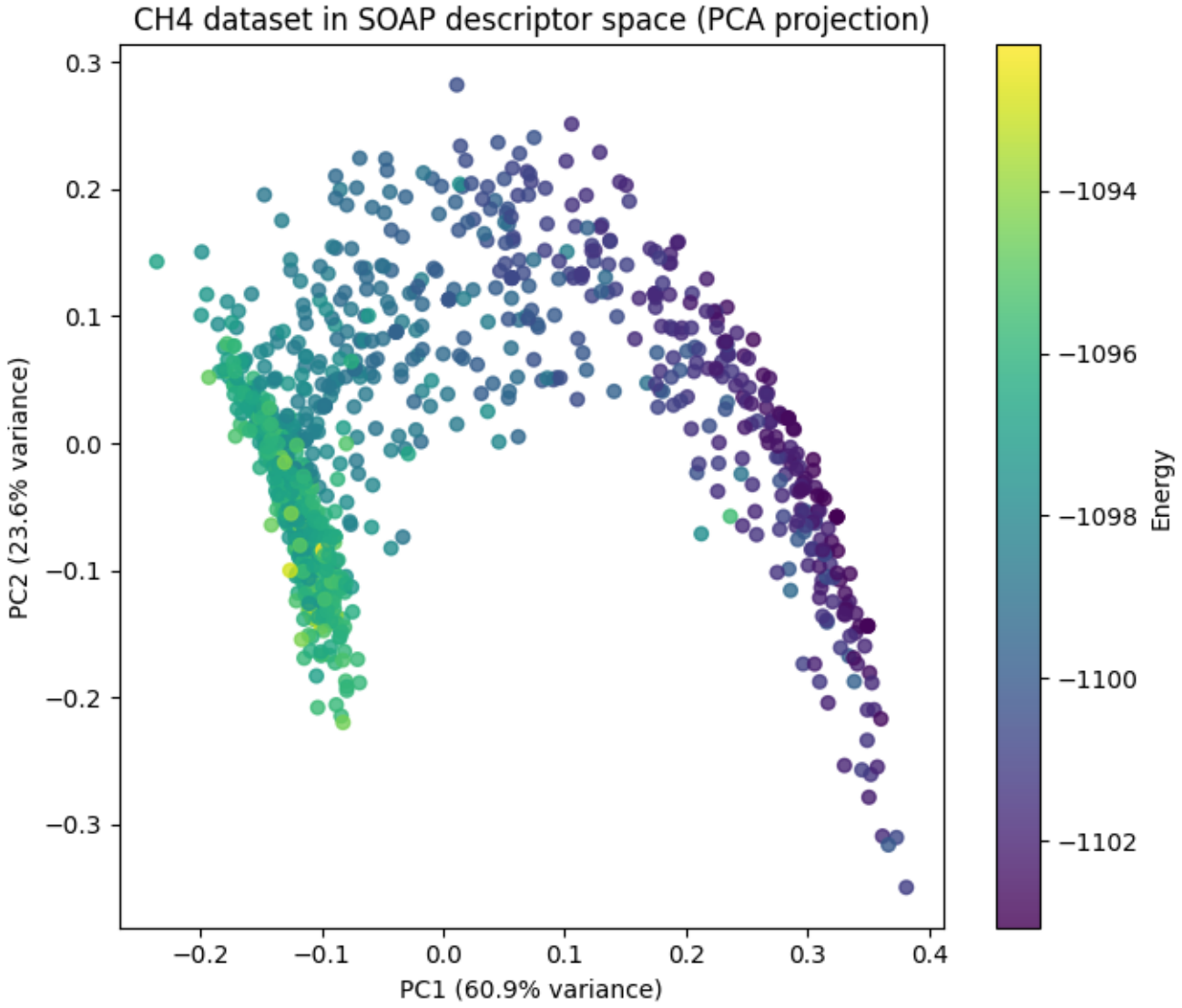
- Active learning framework using GPR.
- Constructs surrogate PES.
- Selects training points in two phases.
- Reduces expensive QM evaluations.



How does ALNEB work in our favour?

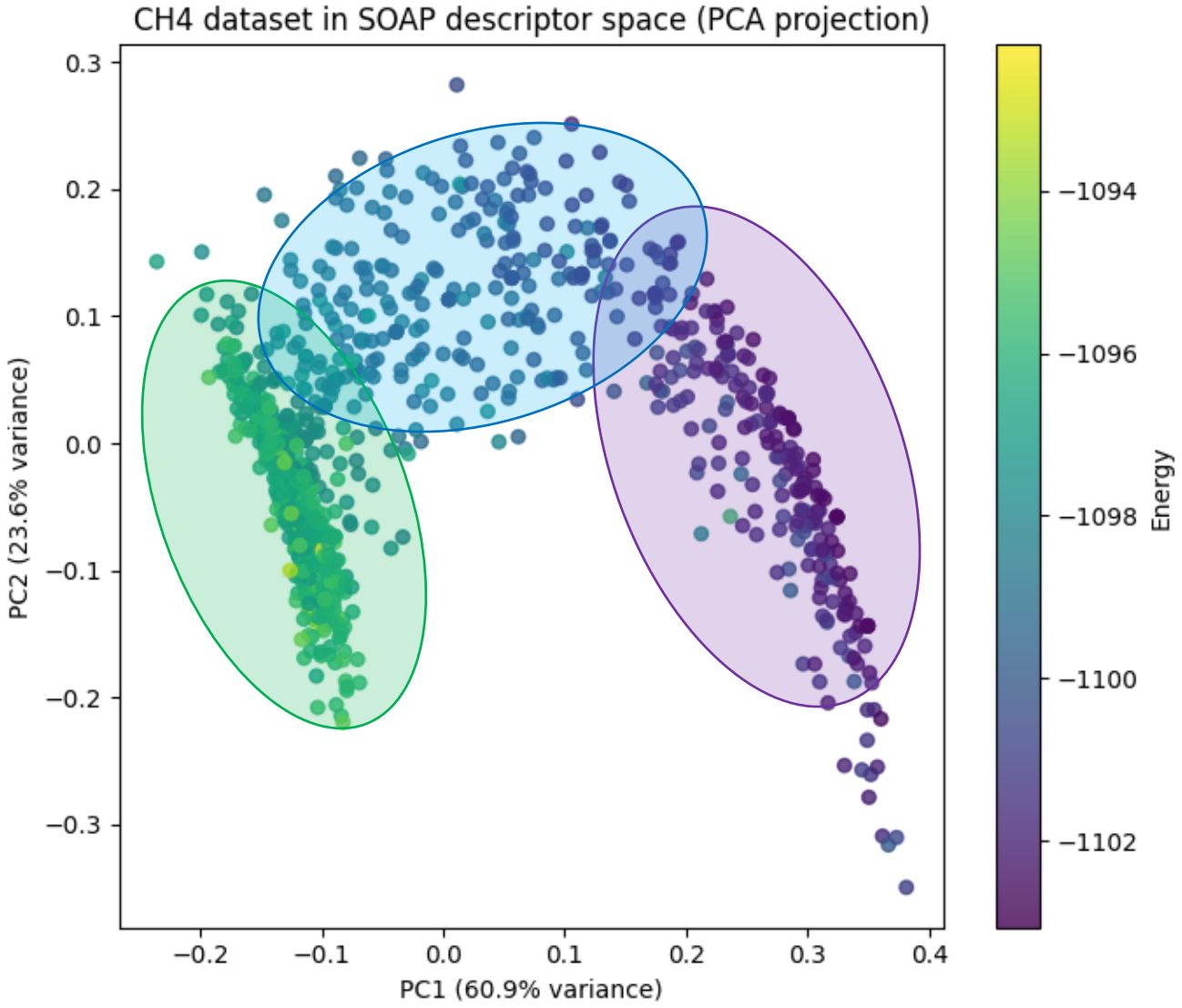


Descriptor Space Visualisation of a Toy Dataset: $CH_4 \rightarrow CH_3 + H$



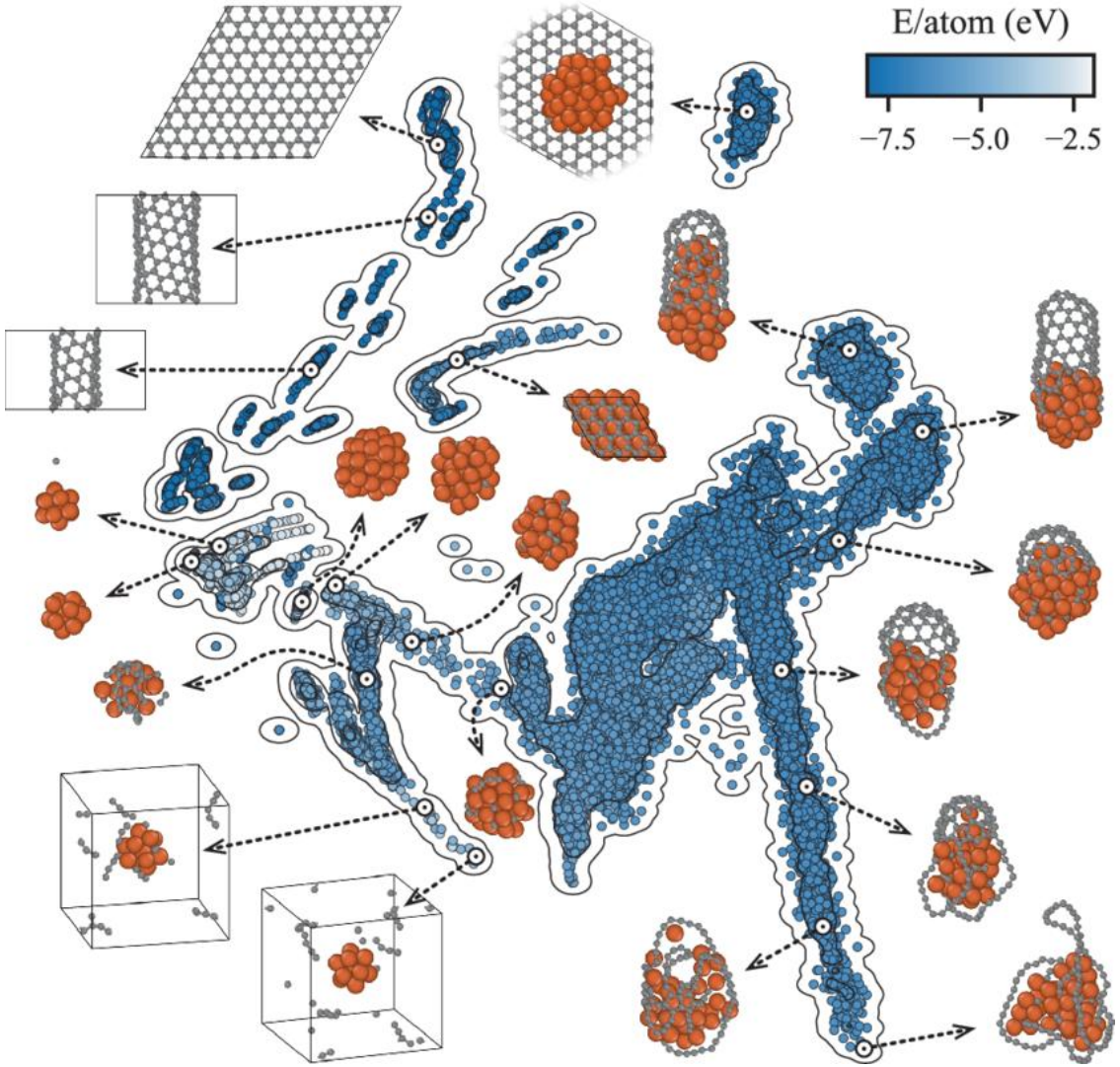
- Generated by random perturbations of the CH_4 bonds.
- psi4 based DFT labelling.

Descriptor Space Visualisation of a Toy Dataset: $CH_4 \rightarrow CH_3 + H$

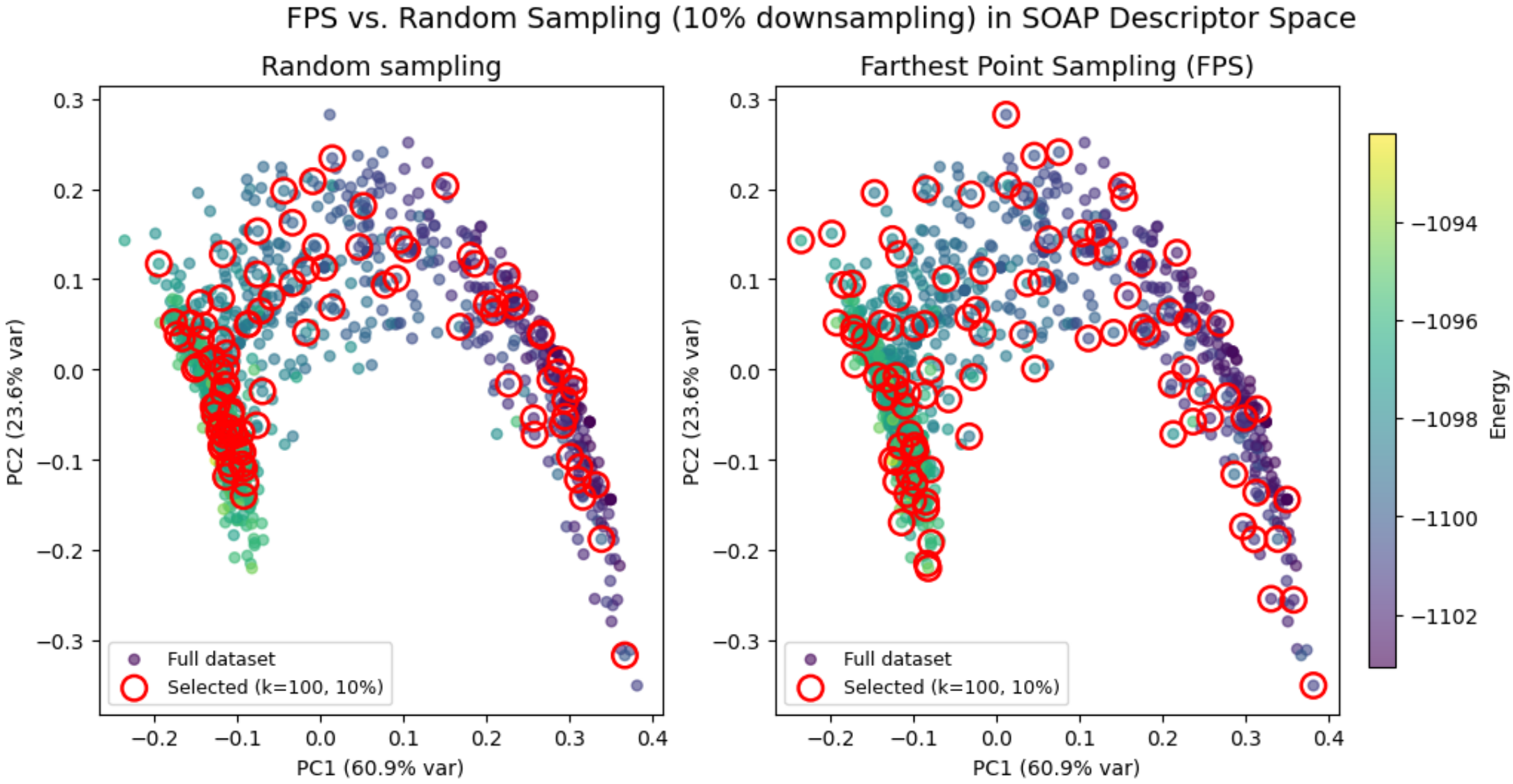


- Generated by random perturbations of the CH_4 bonds.
- psi4 based DFT labelling.
- CH_4 cluster
- CH_3 cluster
- Transition manifold (irrespective of the presence of a defined TS)

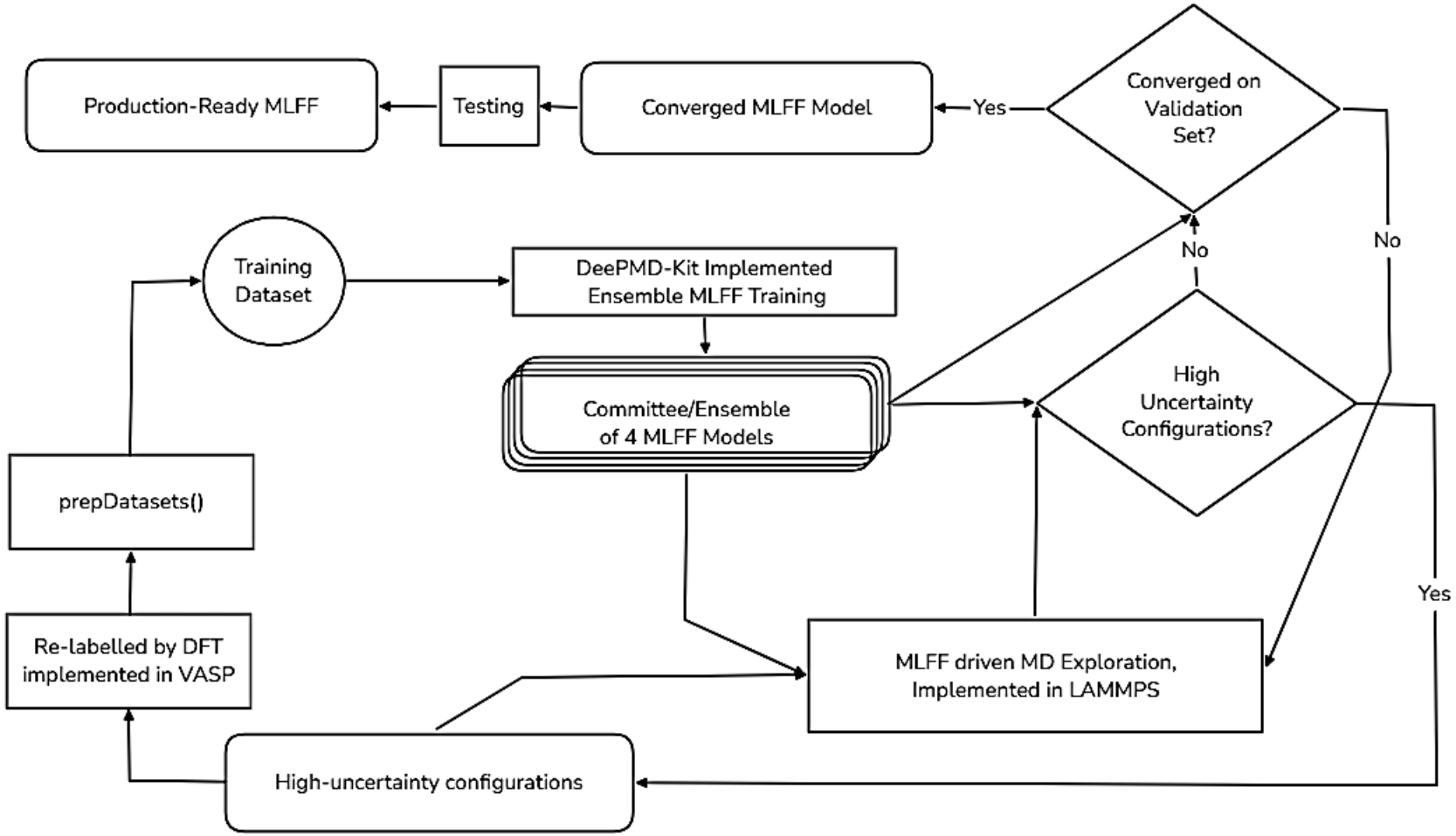
Descriptor Space Visualisation of the Training Dataset: DeepCNT-22



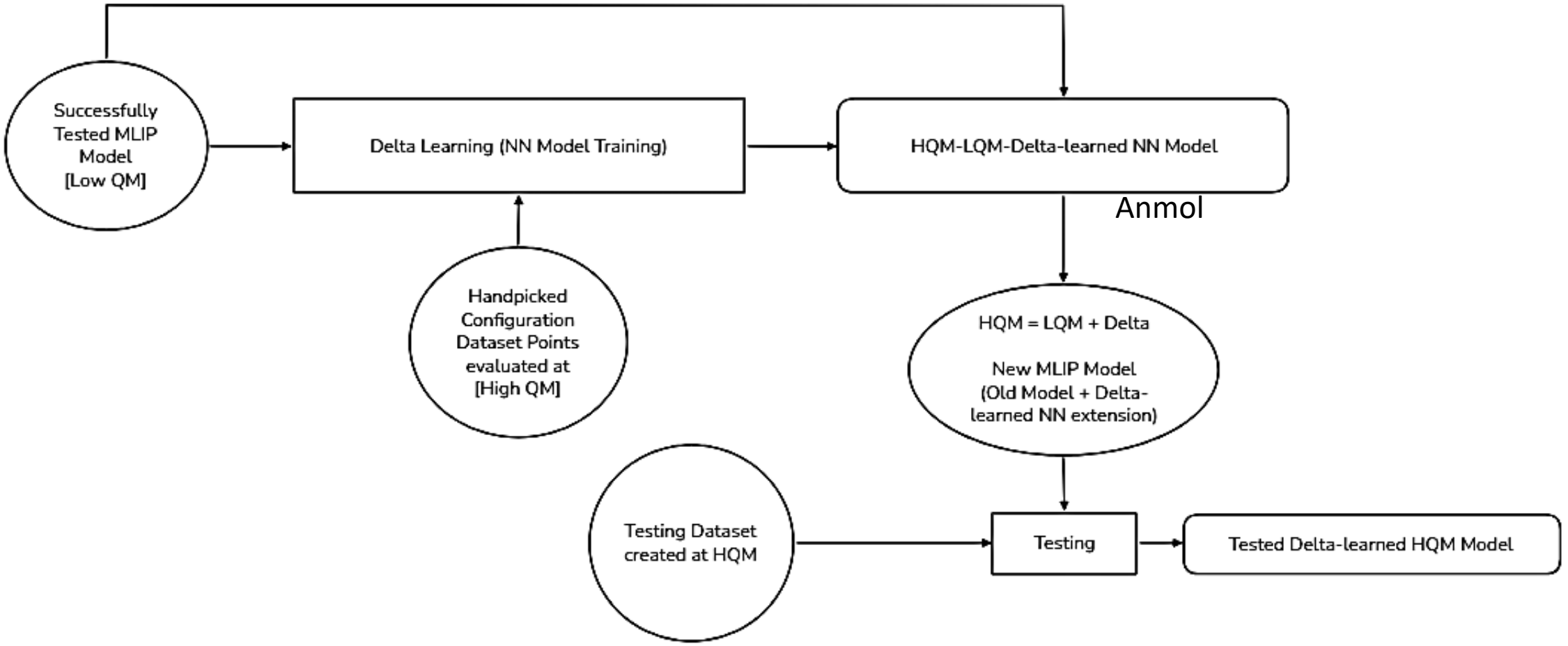
Data Processing: Farthest Point Sampling



MLFF Model Ensemble Training



Implementation of Delta Learning in MLIP Development



Development Milestones

1. Toy Problem: HCN ←→ HNC Isomerisation.
   a. Randomly generated data + single-pass training
   b. ALNEB seed data + single-pass training
   c. ALNEB seed data + AL loop training
   d. ALNEB guided swarm AIMD data + single-pass training
   e. ALNEB guided swarm AIMD data + AL loop training
   f. Generalisation and HPC-compatible script preparation

Anmol

2. Toy Problem: CH_4 → CH_3 + H
   a. Similar milestones as the HCN toy problem

3. Result Benchmarking: GAP-20

4. Result Benchmarking: DeepCNT-22

5. Production: Pure Metal + Alloy (Cu-In) MLIP

6. Production: CNT Growth on Molten Alloy Nanoparticle MLIP

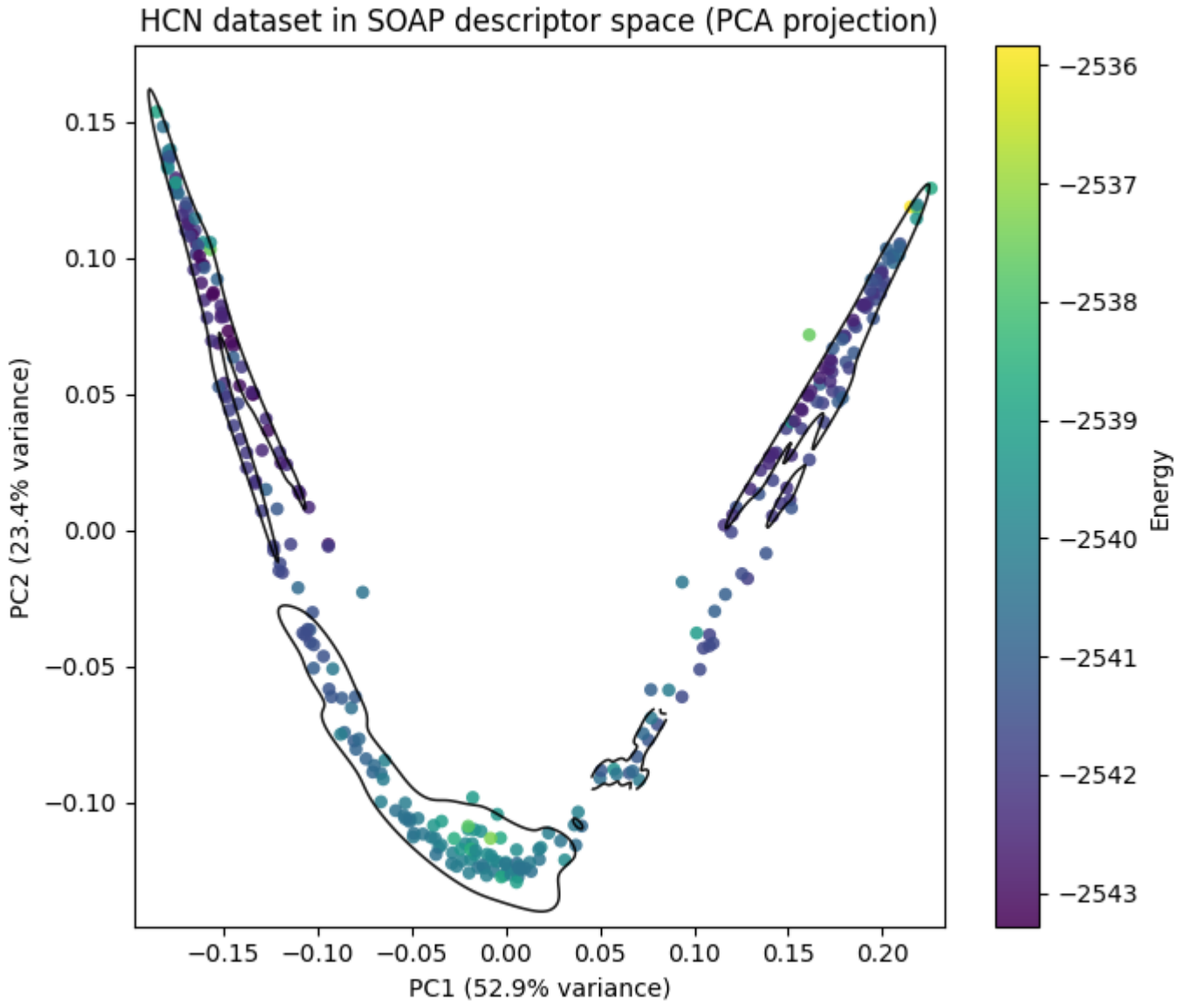
Part 4:

Toy Problems

HCN ←→ HNC Isomerisation

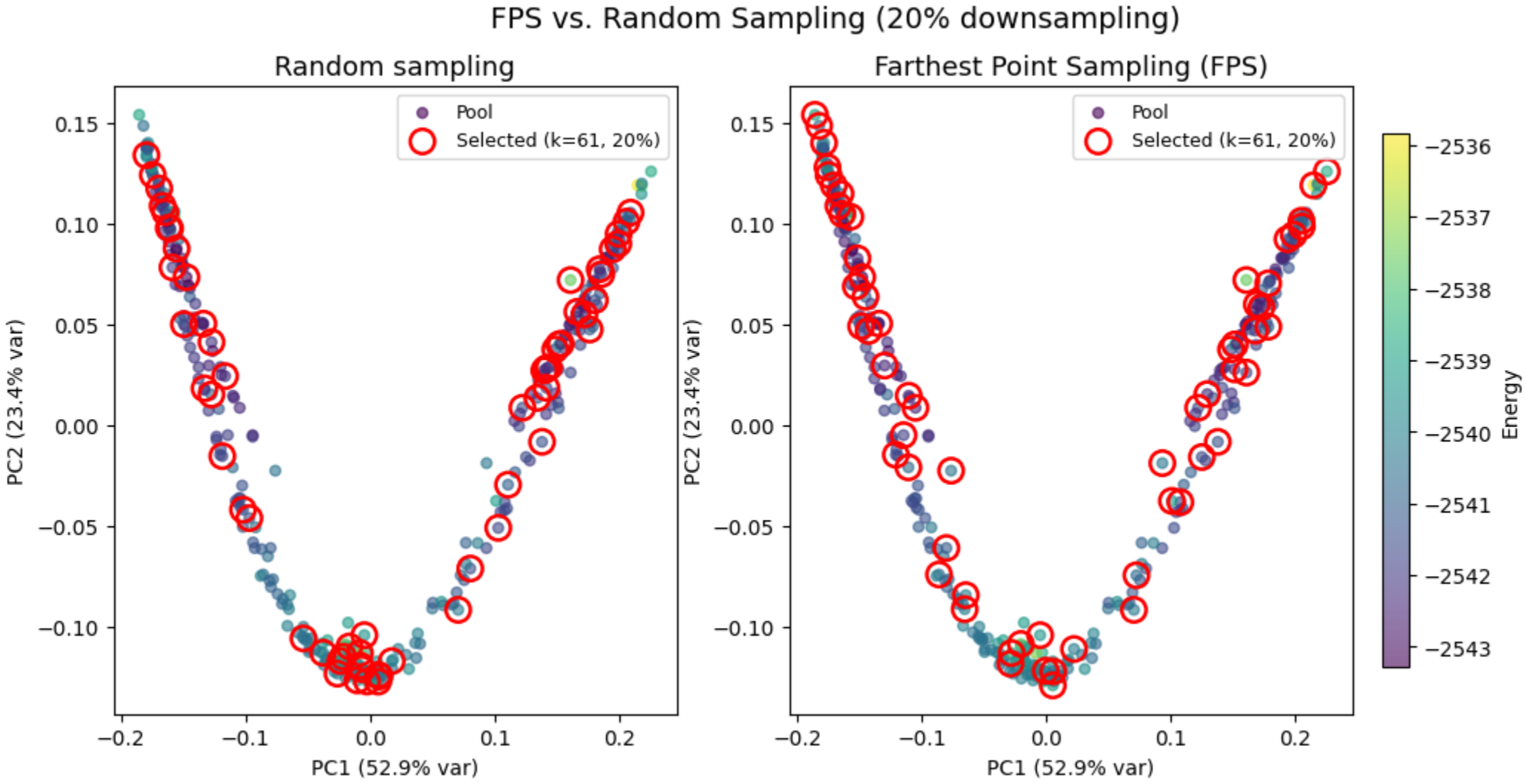
CH_4 Dissociation

Toy Problem: HCN ←→ HNC Isomerisation

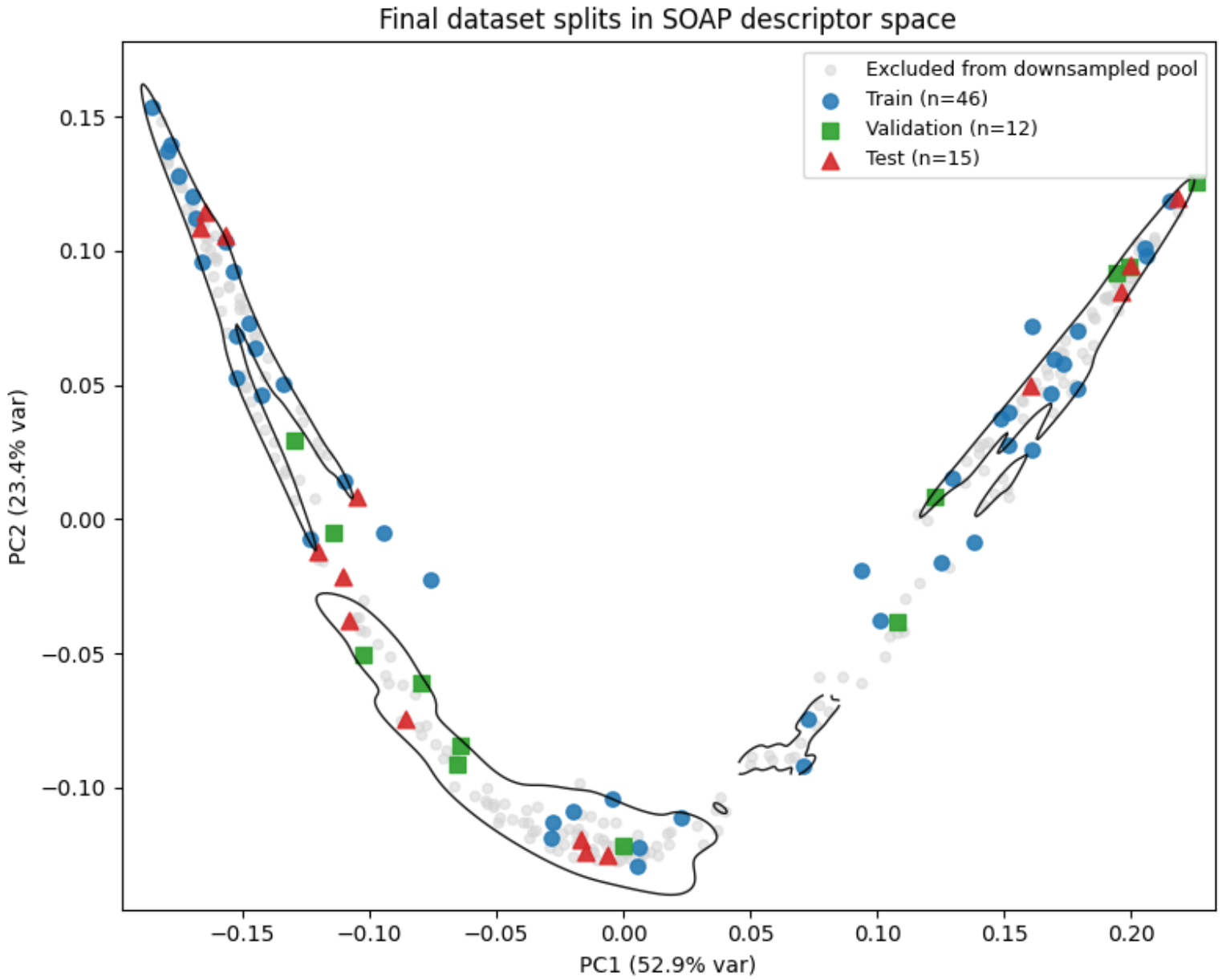


- Generated by random perturbations of the HCN bonds.
- psi4 based DFT labelling.

Toy Problem: HCN ←→ HNC Isomerisation

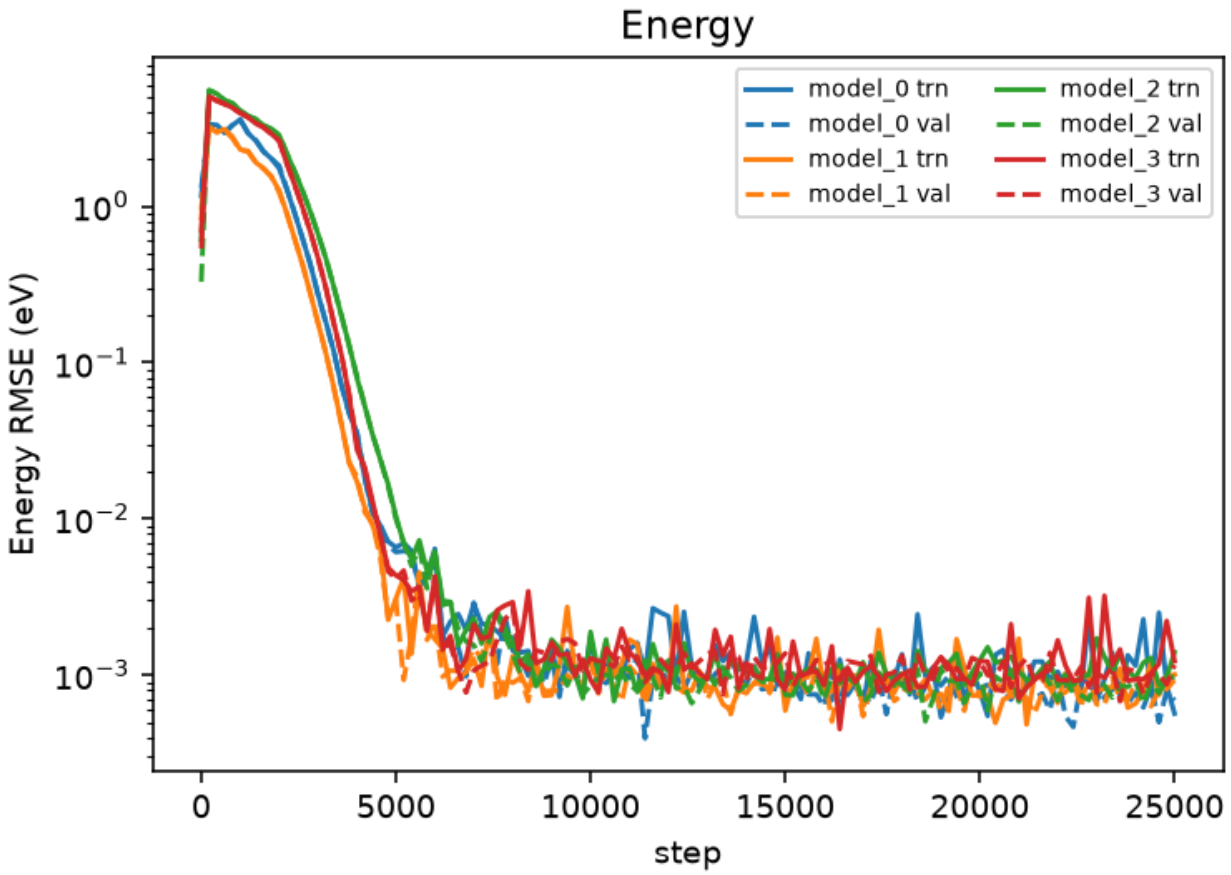


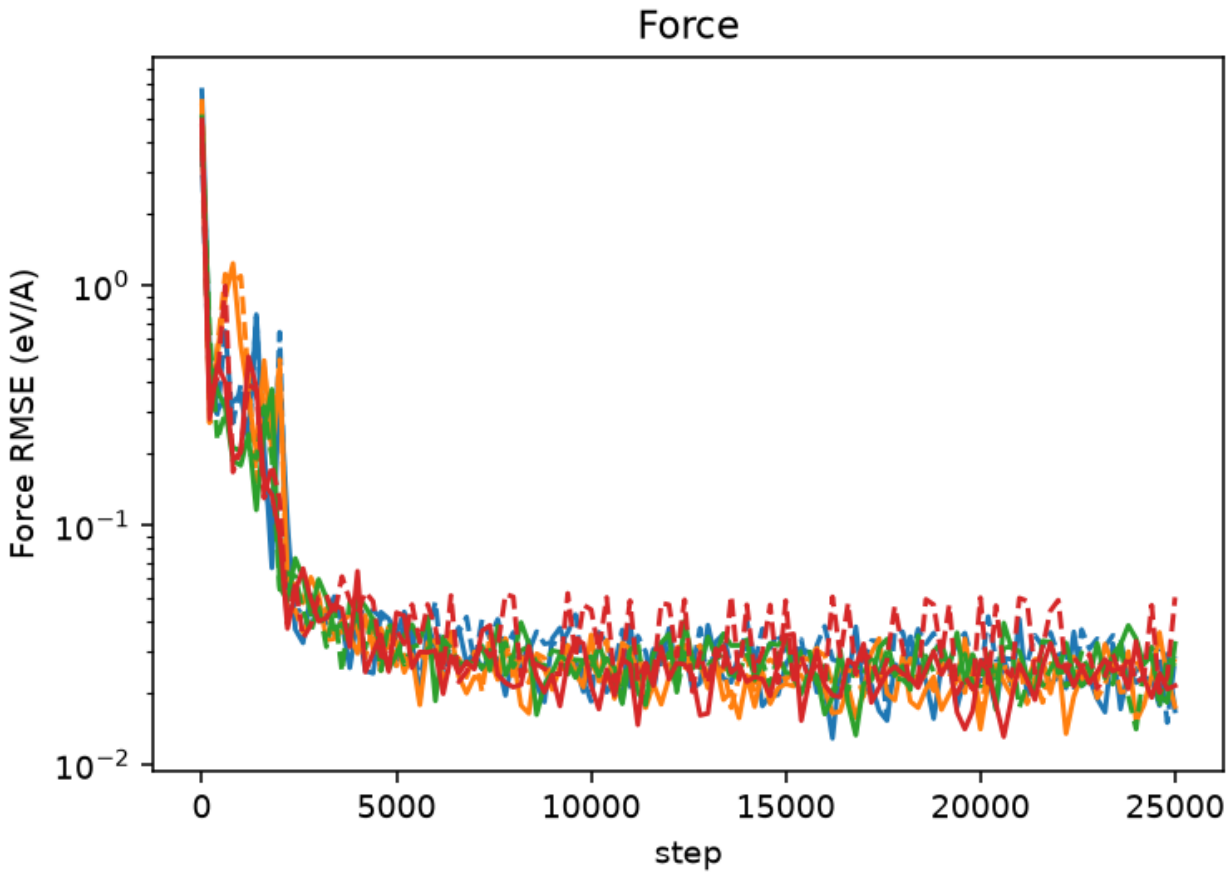
Toy Problem: HCN ←→ HNC Isomerisation



Toy Problem: HCN ←→ HNC Isomerisation

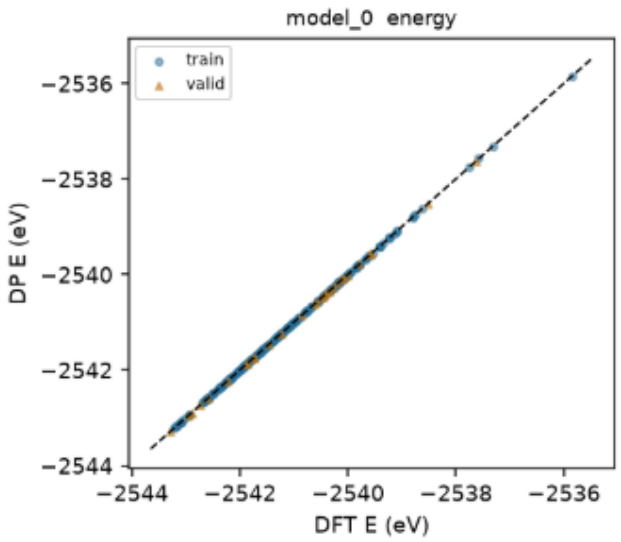
Learning Curves & RMSE values

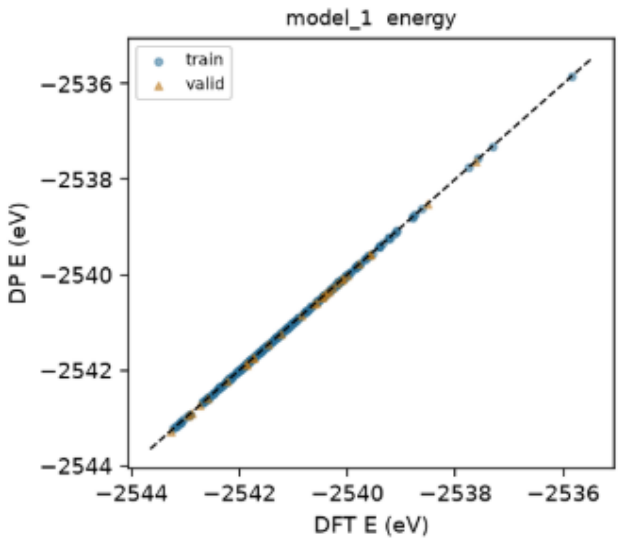


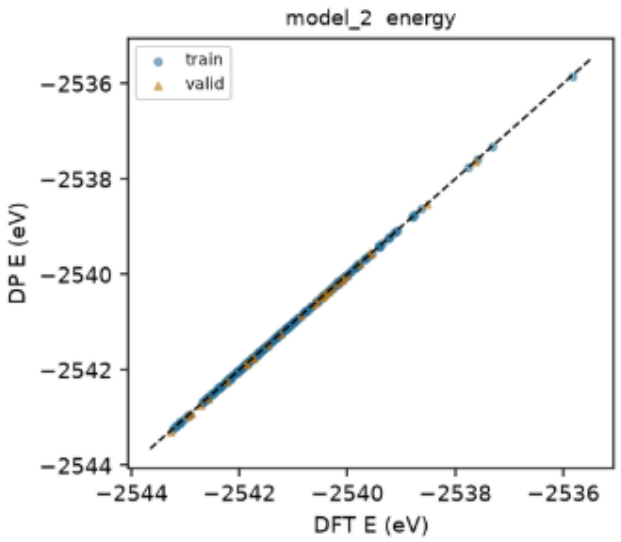


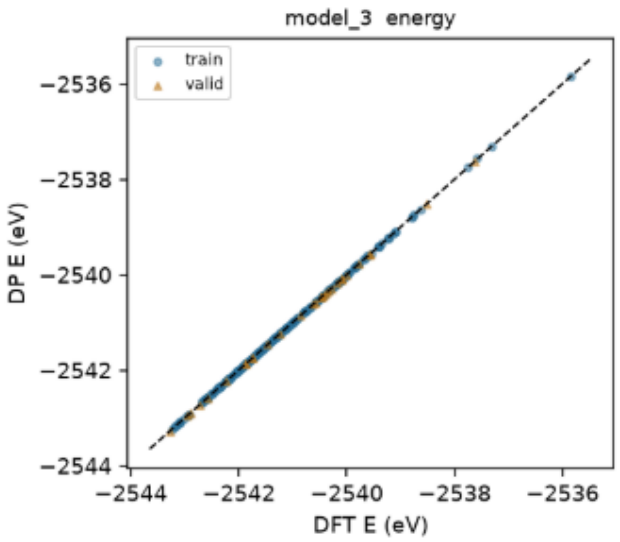
Toy Problem: HCN ←→ HNC Isomerisation

Energy Parity Plots: Training and Validation



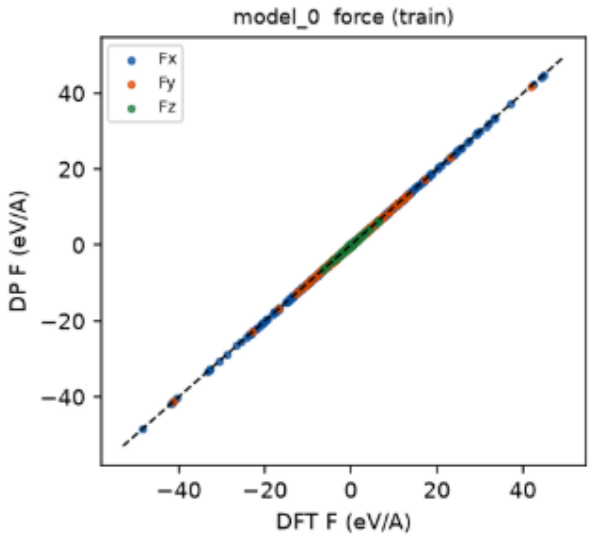


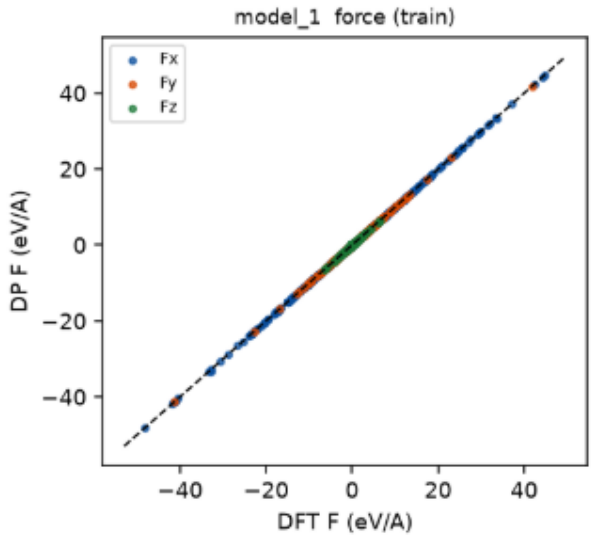


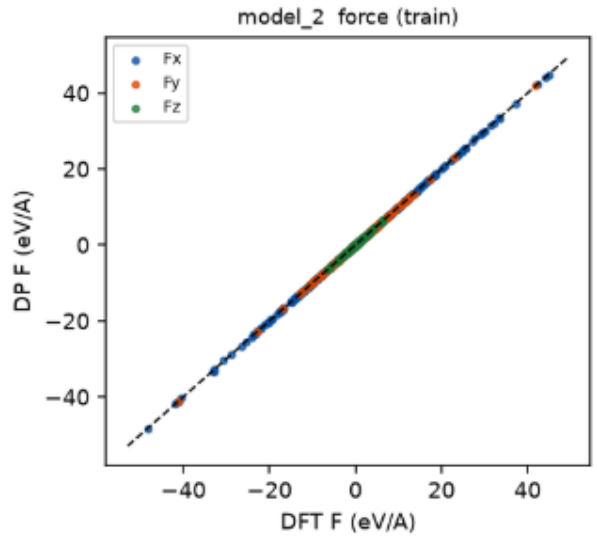


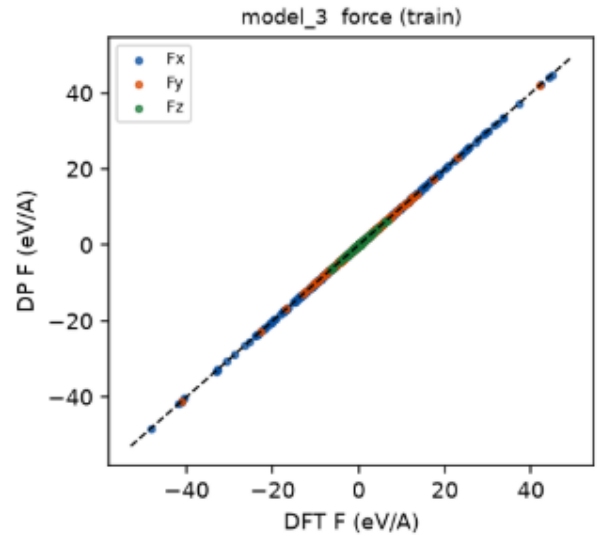
Toy Problem: HCN ←→ HNC Isomerisation

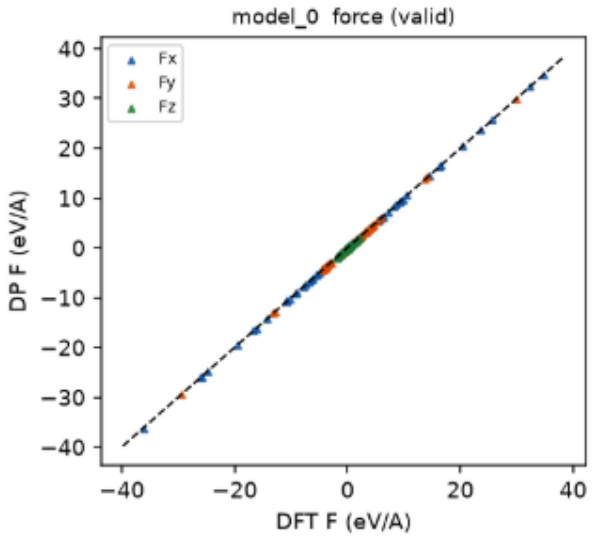
Force Parity Plots: Training and Validation

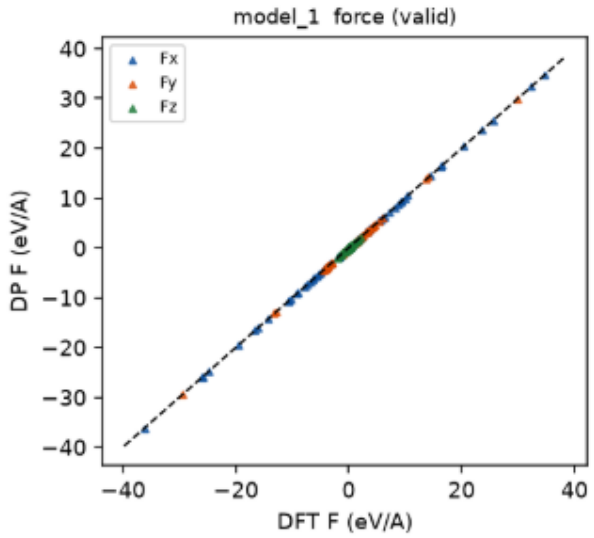


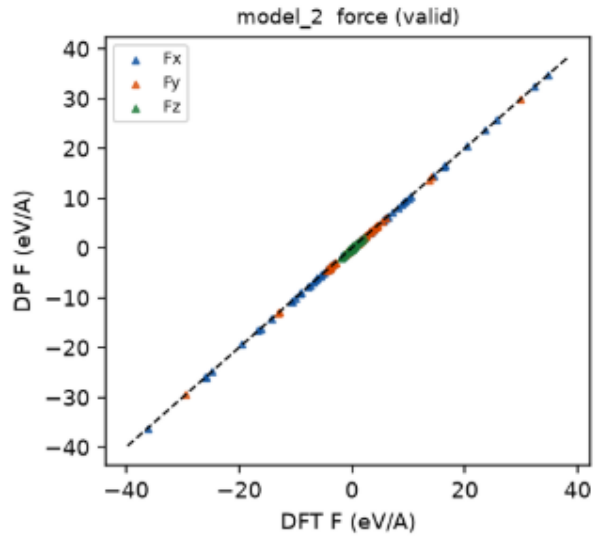


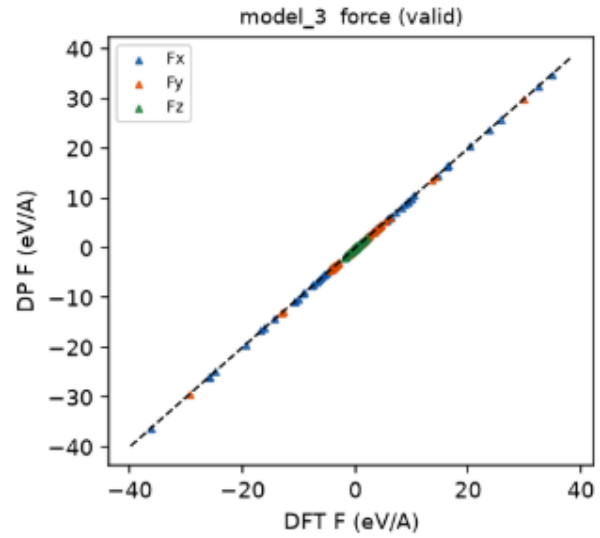






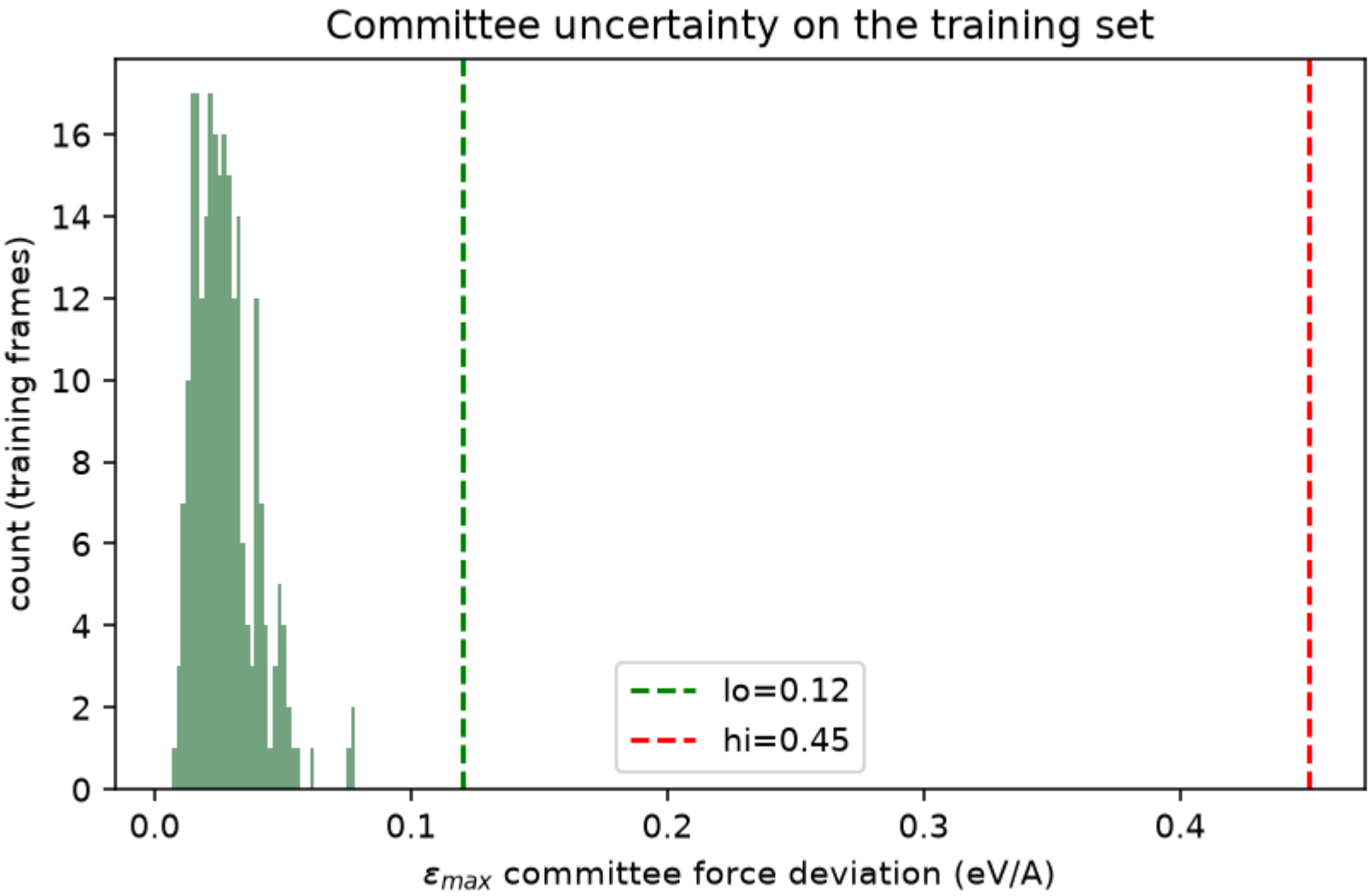






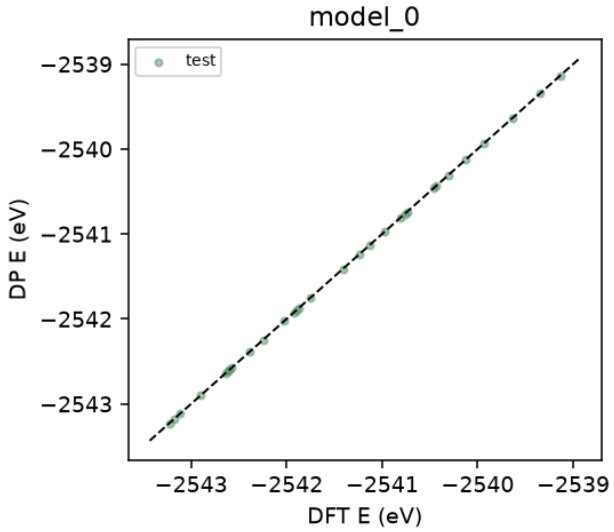
Toy Problem: HCN ←→ HNC Isomerisation

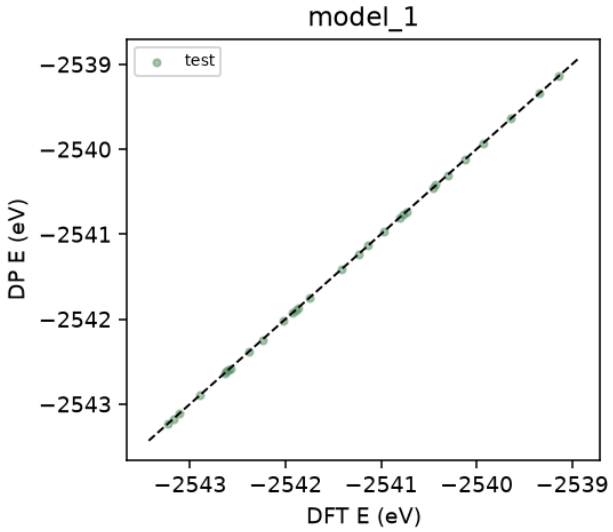
Committee Uncertainty Histogram

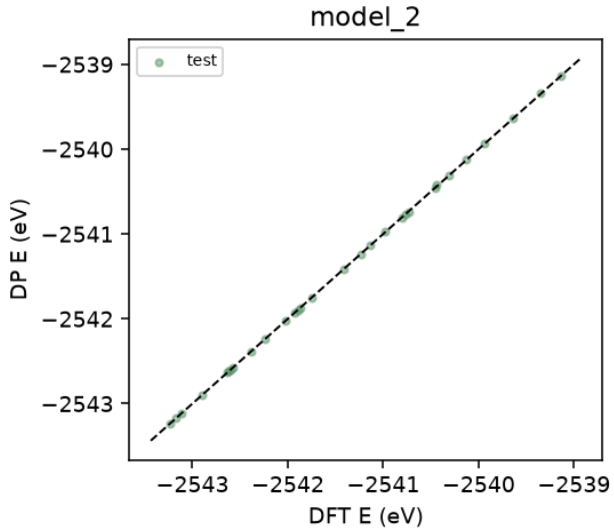


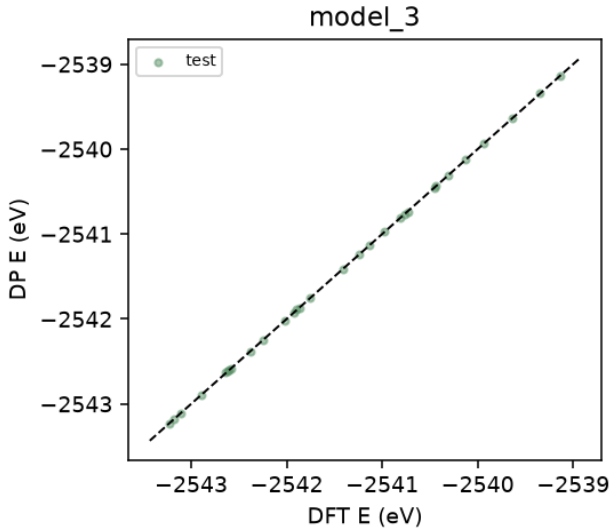
Toy Problem: HCN ←→ HNC Isomerisation

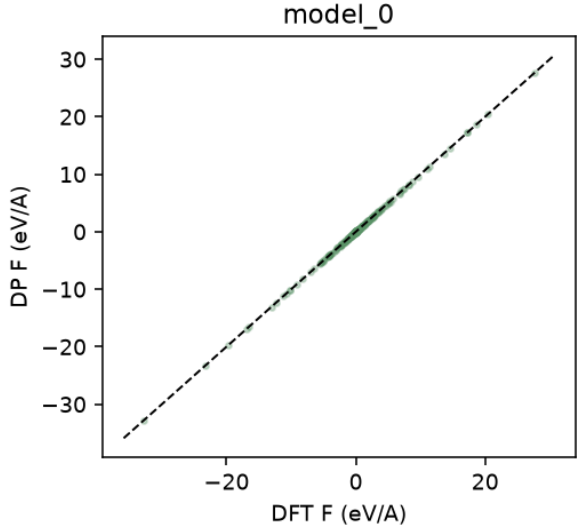
Parity Plots: Testing

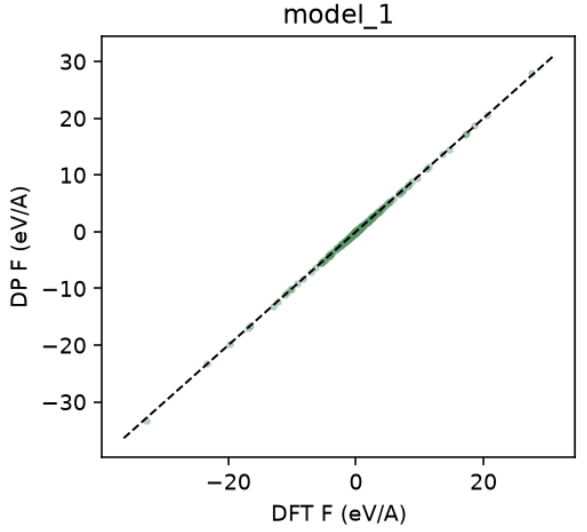


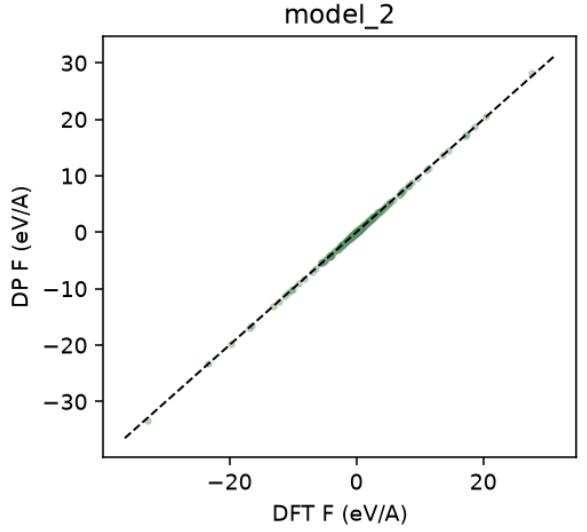


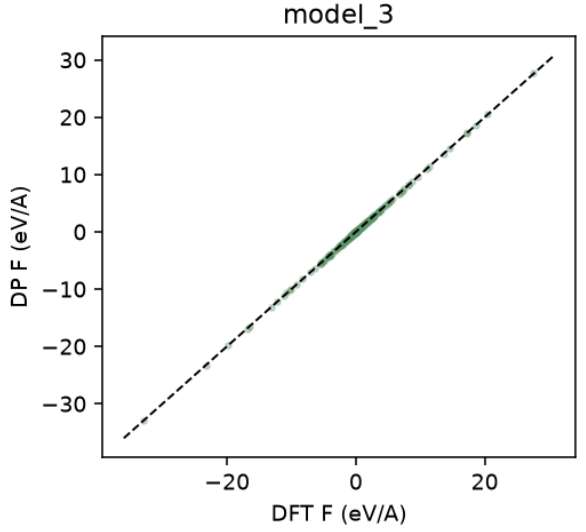






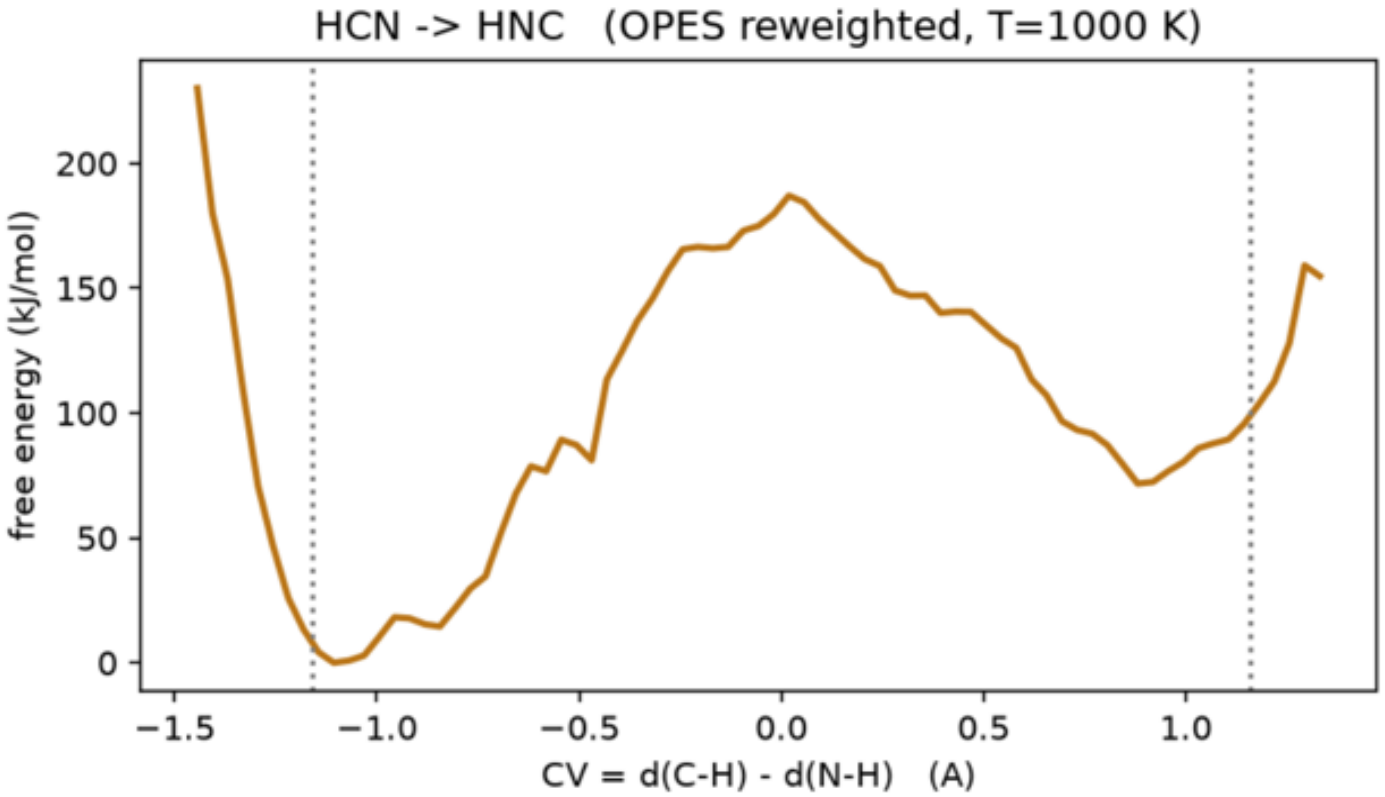




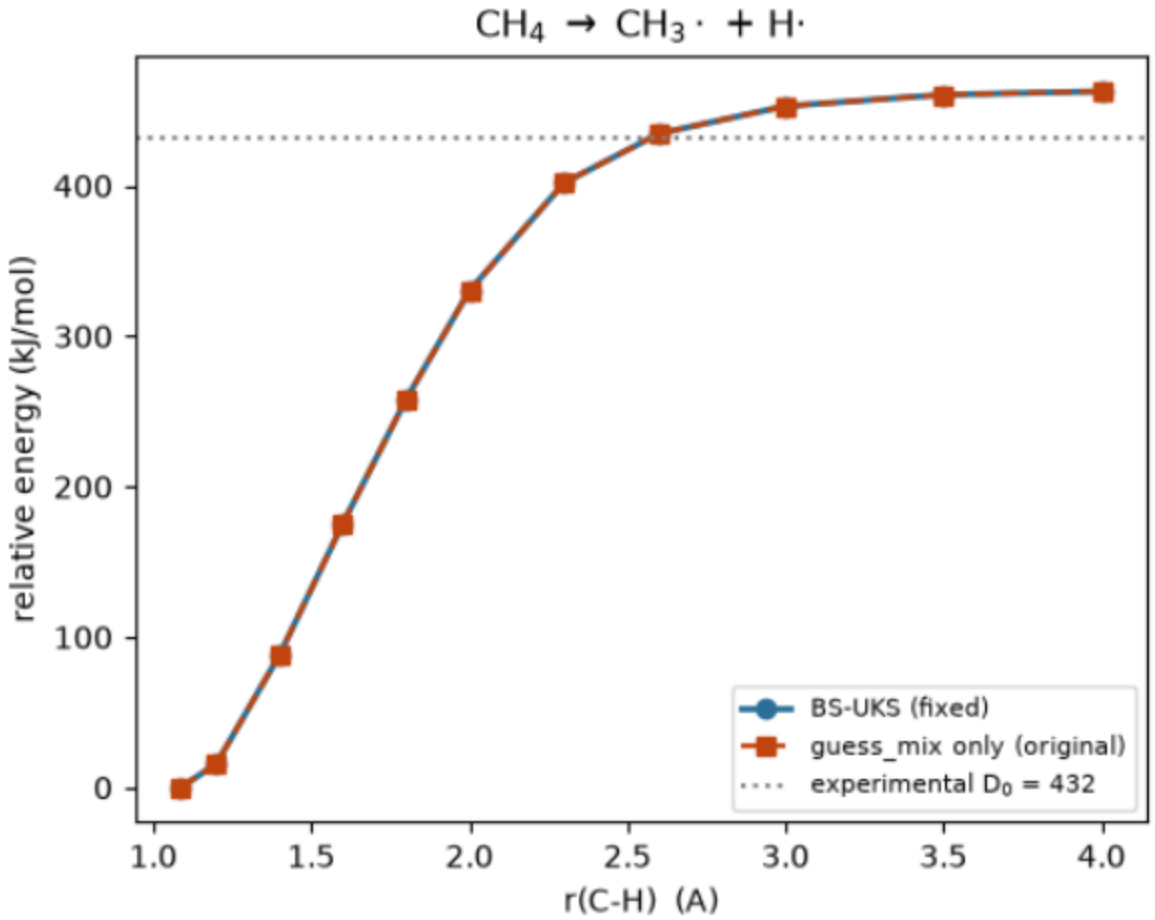


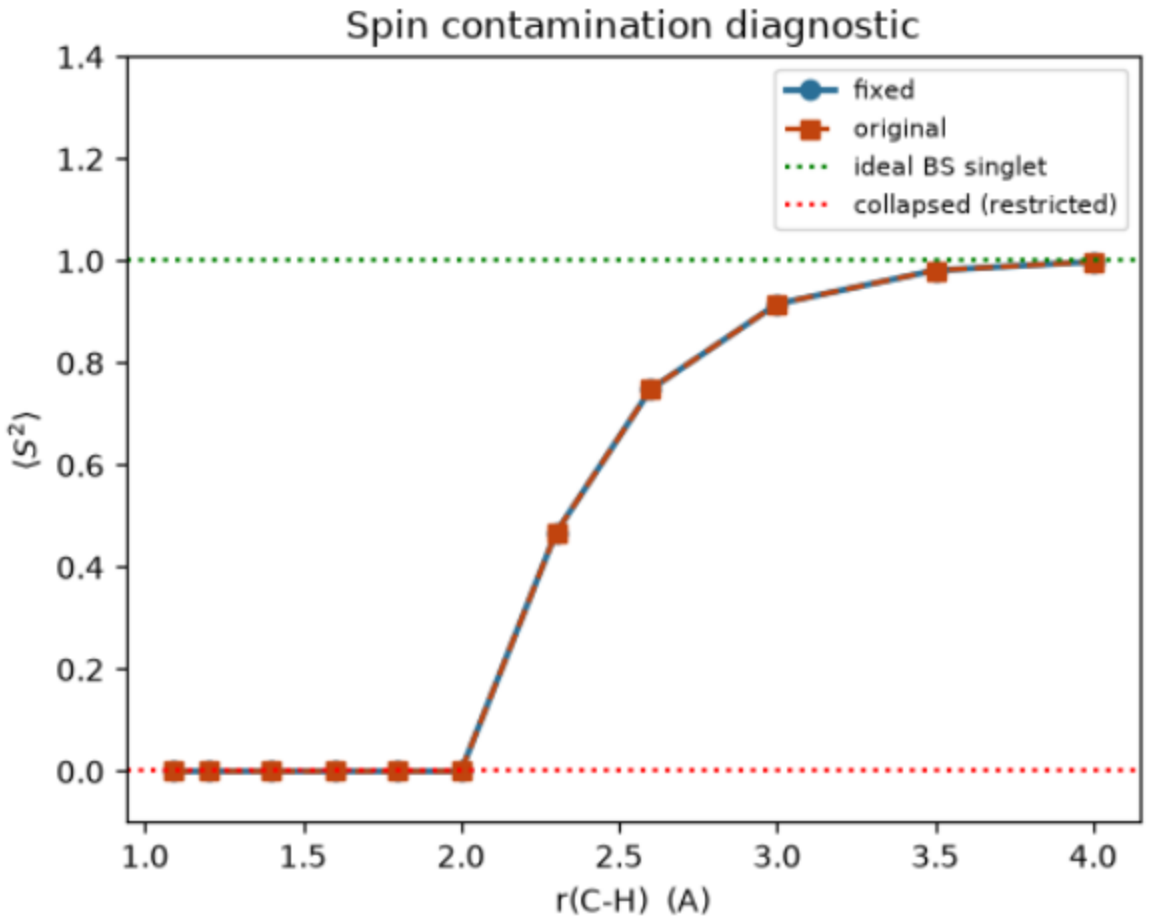
Toy Problem: HCN ←→ HNC Isomerisation

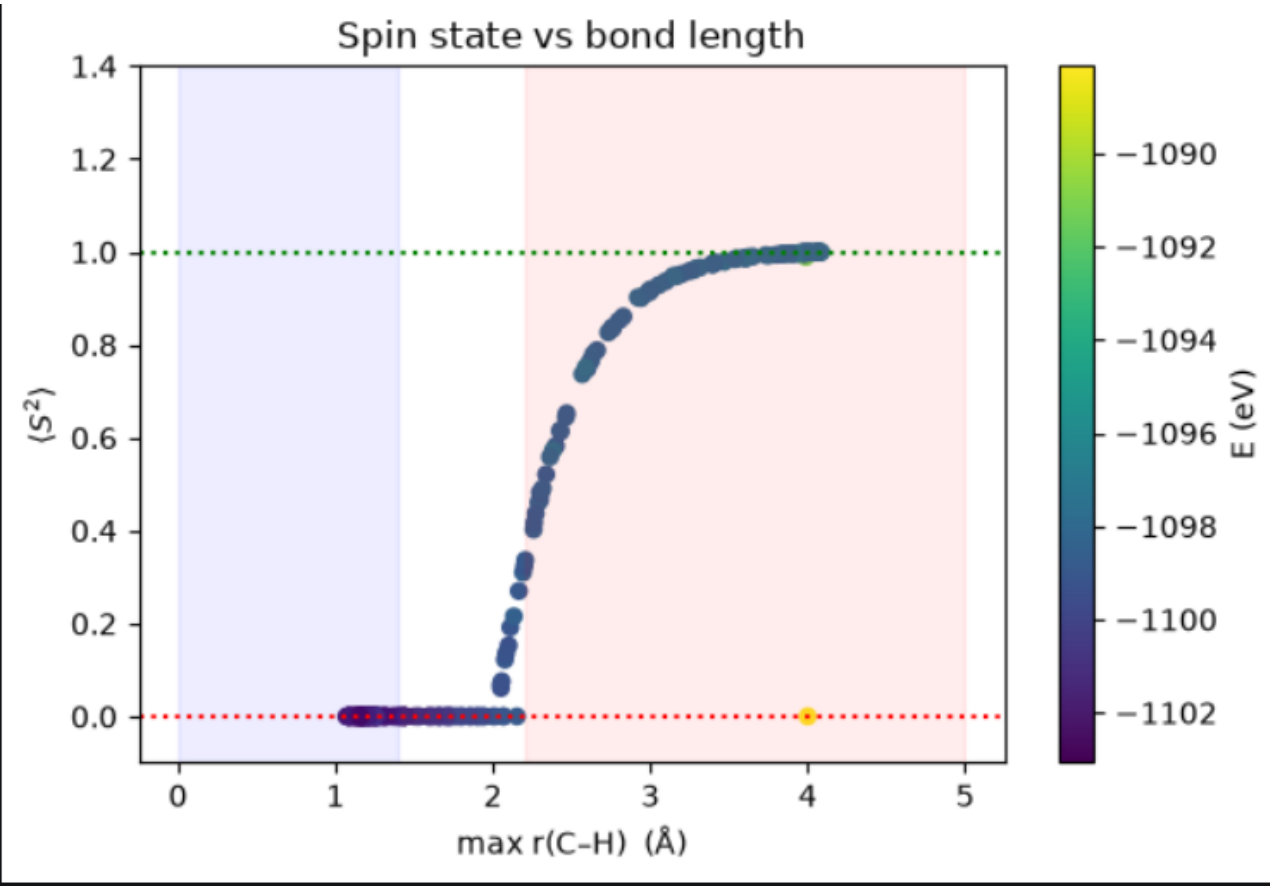
Metadynamics Run

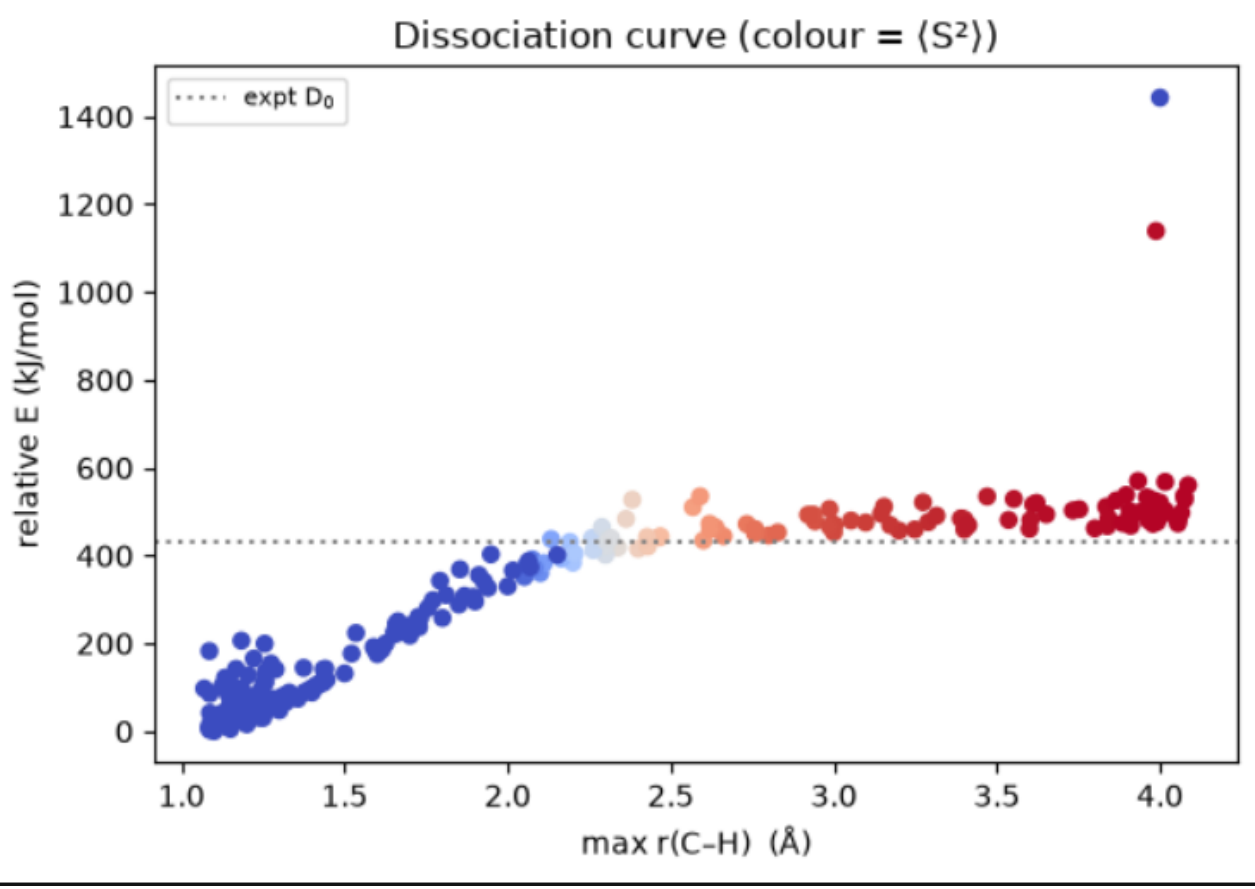


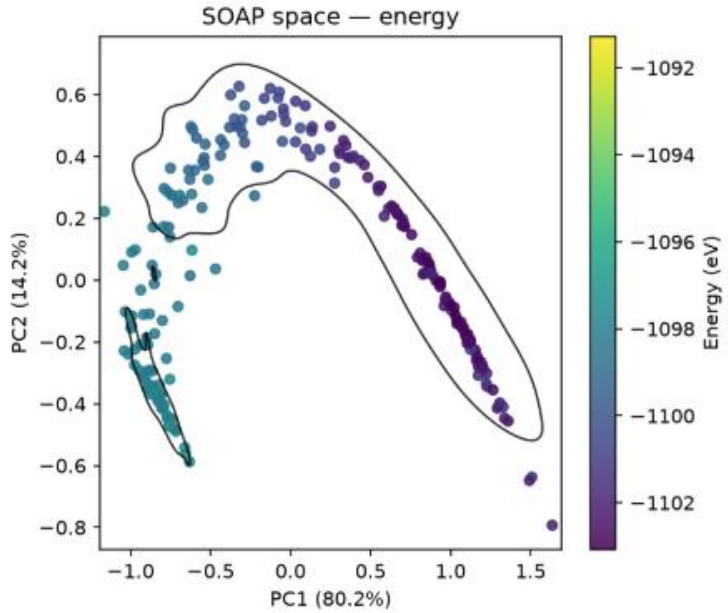
The MLIP reports the activation energy barrier for the forward reaction as 187 kJ/mol. This matches the literature benchmarked value which lies in the range of 180 kJ/mol - 190 kJ/mol.

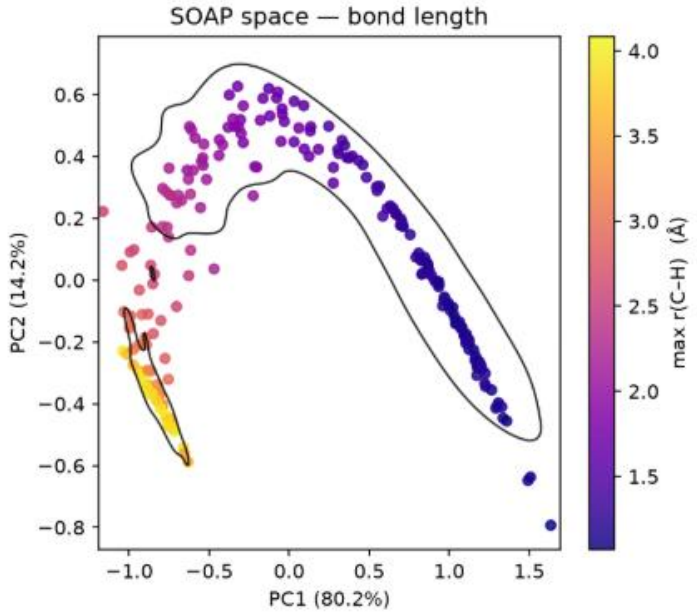

$CH_4 \rightarrow CH_3\cdot + H\cdot$
relative energy (kJ/mol)
0
100
200
300
400
1.0
1.5
2.0
2.5
3.0
3.5
4.0
r(C-H) (A)
BS-UKS (fixed)
guess_mix only (original)
experimental D_0 = 432

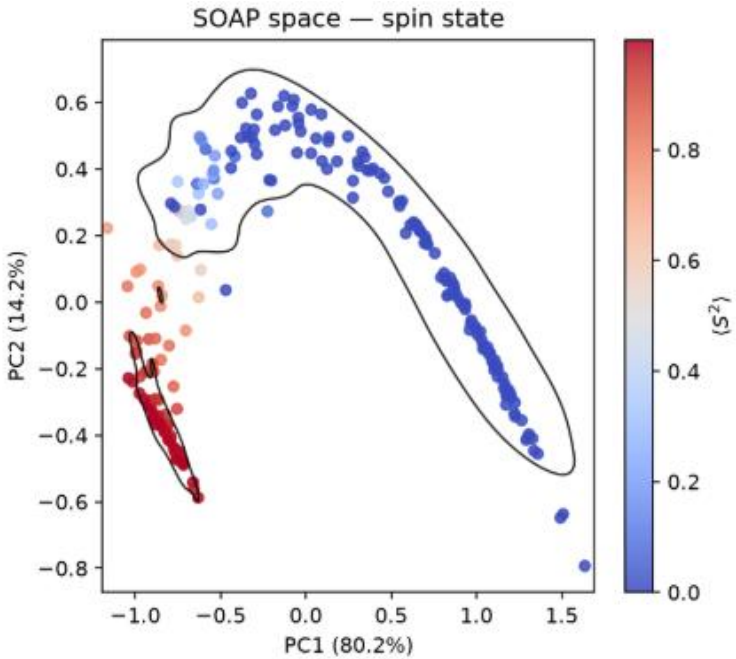

Spin contamination diagnostic
$\langle S^2 \rangle$
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.0
1.5
2.0
2.5
3.0
3.5
4.0
r(C-H) (A)
fixed
original
ideal BS singlet
collapsed (restricted)

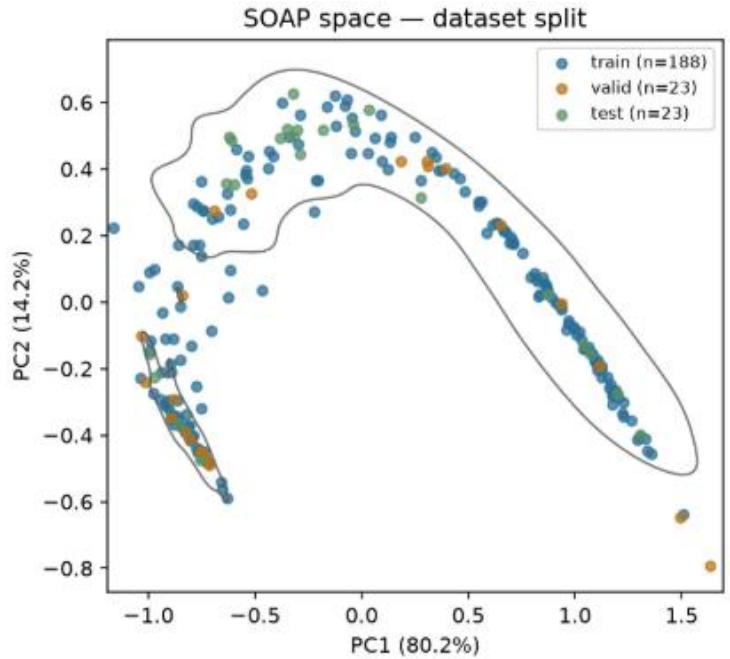

Spin state vs bond length
⟨S²⟩
max r(C–H) (Å)
E (eV)

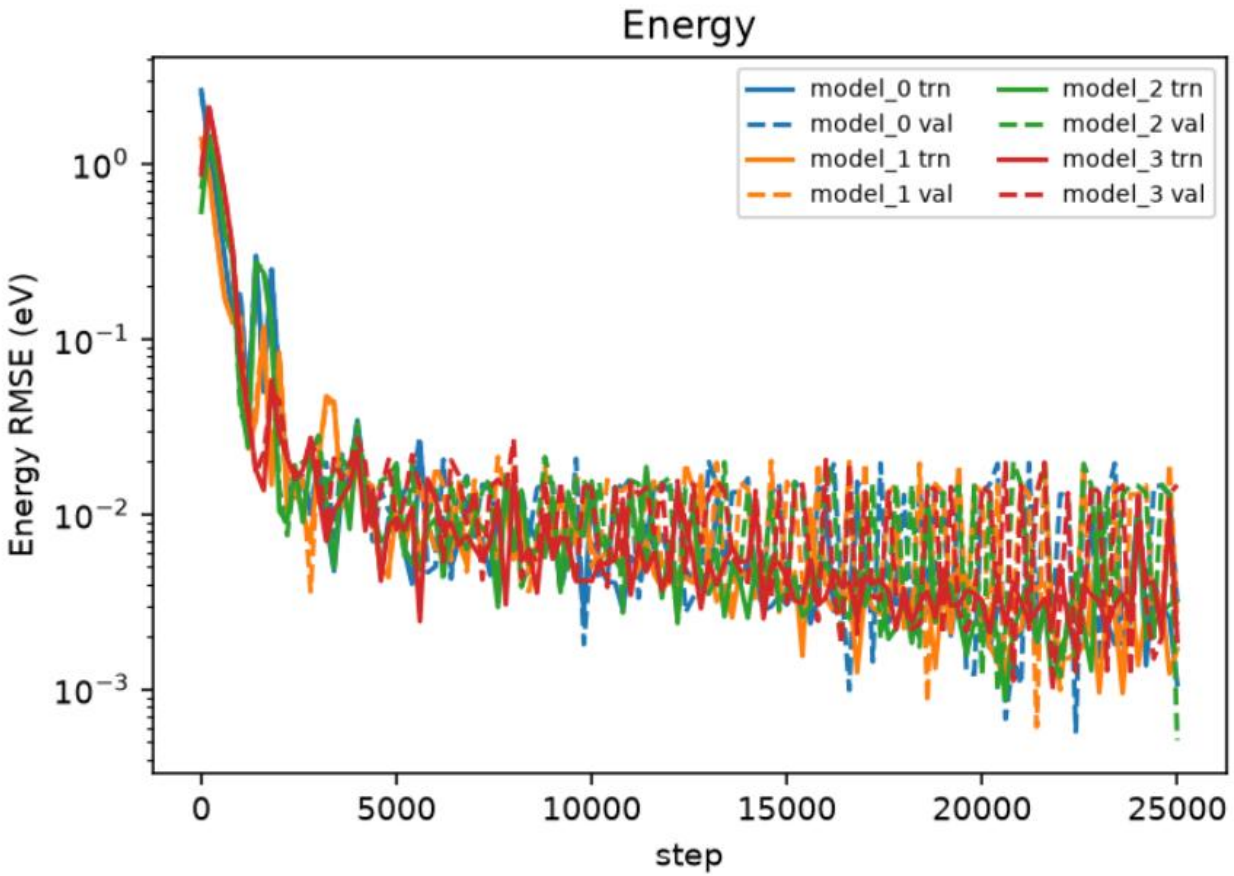

Dissociation curve (colour = ⟨S²⟩)
expt D₀
relative E (kJ/mol)
max r(C–H) (Å)

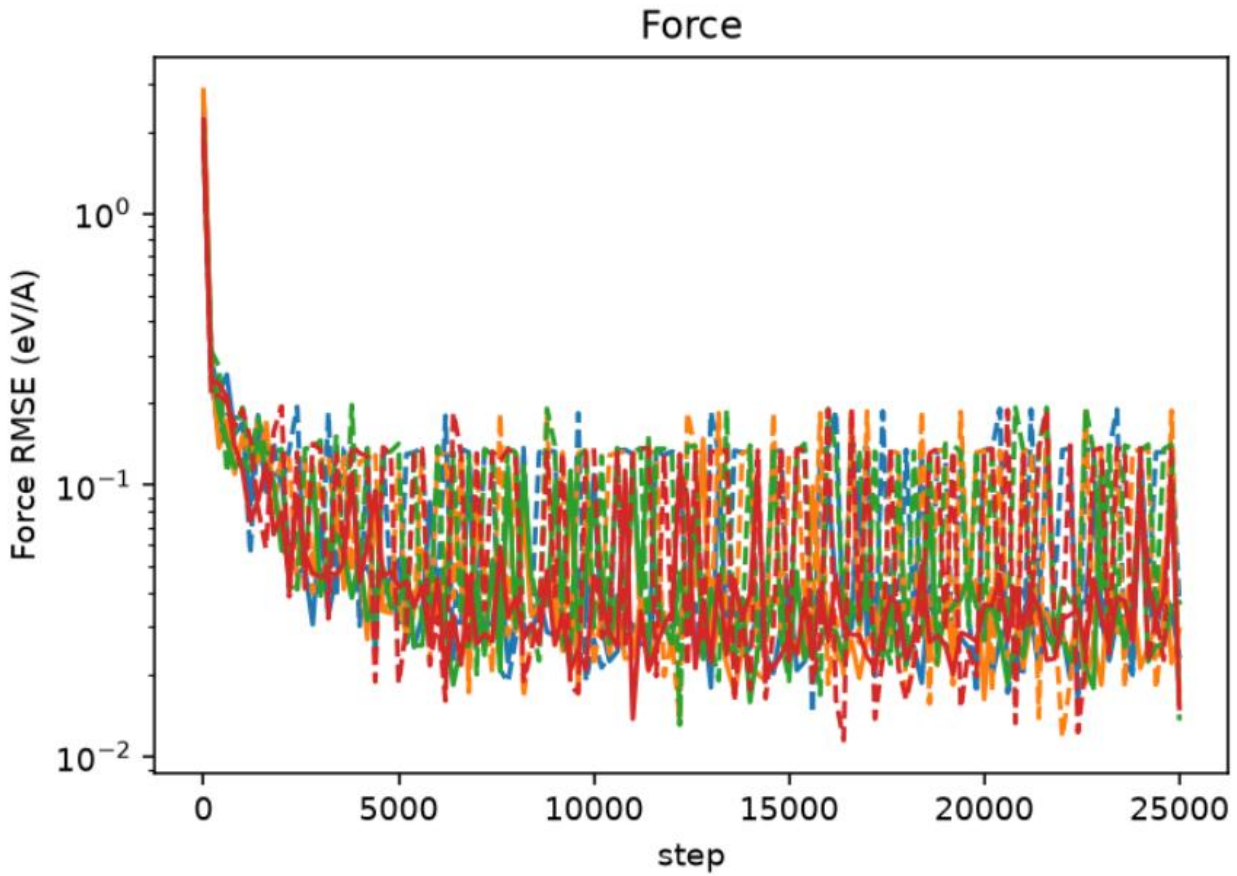

SOAP space — energy
PC1 (80.2%)
PC2 (14.2%)
Energy (eV)
−1092
−1094
−1096
−1098
−1100
−1102

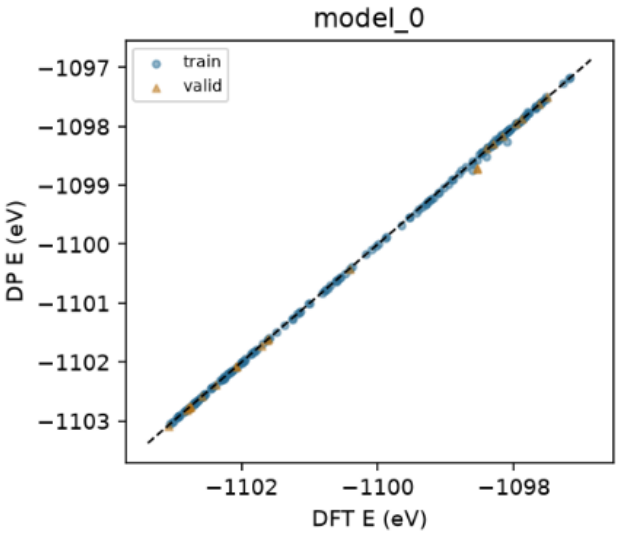

SOAP space — bond length
PC1 (80.2%)
PC2 (14.2%)
max r(C–H) (Å)
4.0
3.5
3.0
2.5
2.0
1.5

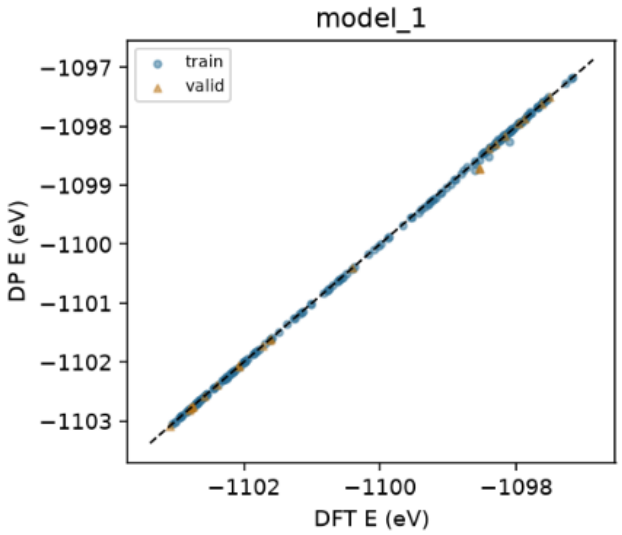

SOAP space — spin state
PC1 (80.2%)
PC2 (14.2%)
⟨S²⟩
0.8
0.6
0.4
0.2
0.0

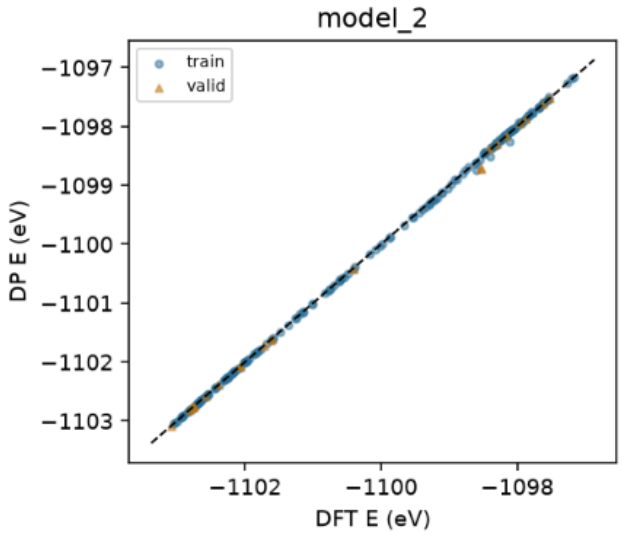

SOAP space — dataset split
train (n=188)
valid (n=23)
test (n=23)
PC1 (80.2%)
PC2 (14.2%)

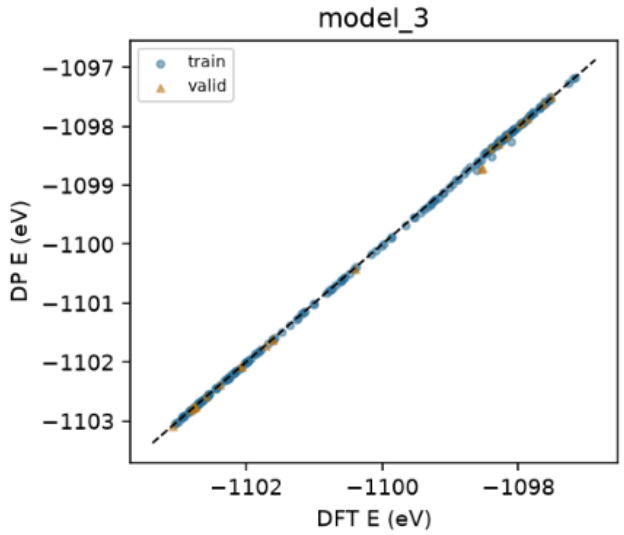

Energy
model_0 trn
model_0 val
model_1 trn
model_1 val
model_2 trn
model_2 val
model_3 trn
model_3 val
Energy RMSE (eV)
10^{0}
10^{-1}
10^{-2}
10^{-3}
0
5000
10000
15000
20000
25000
step

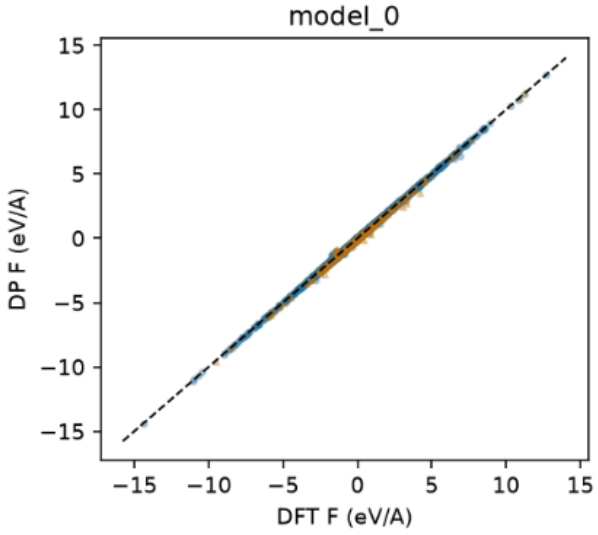

Force
Force RMSE (eV/A)
10^{0}
10^{-1}
10^{-2}
0
5000
10000
15000
20000
25000
step

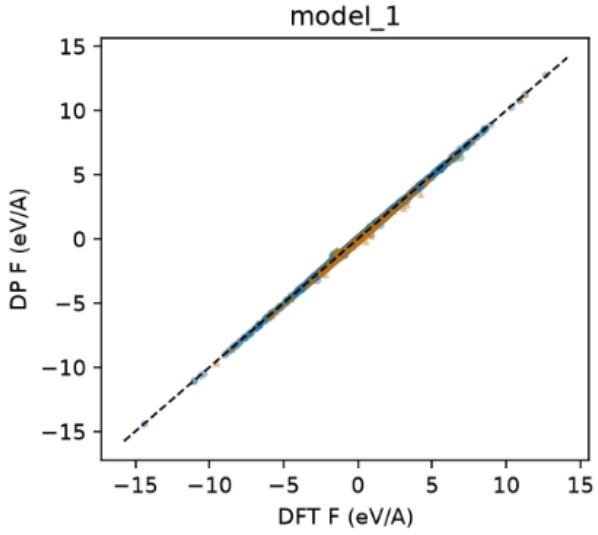

model_0
train
valid
DP E (eV)
DFT E (eV)
−1097
−1098
−1099
−1100
−1101
−1102
−1103
−1102
−1100
−1098

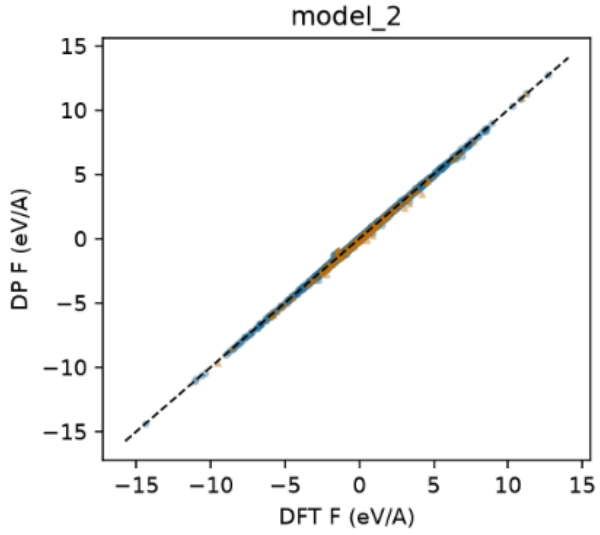

model_1
train
valid
DP E (eV)
DFT E (eV)
−1097
−1098
−1099
−1100
−1101
−1102
−1103
−1102
−1100
−1098

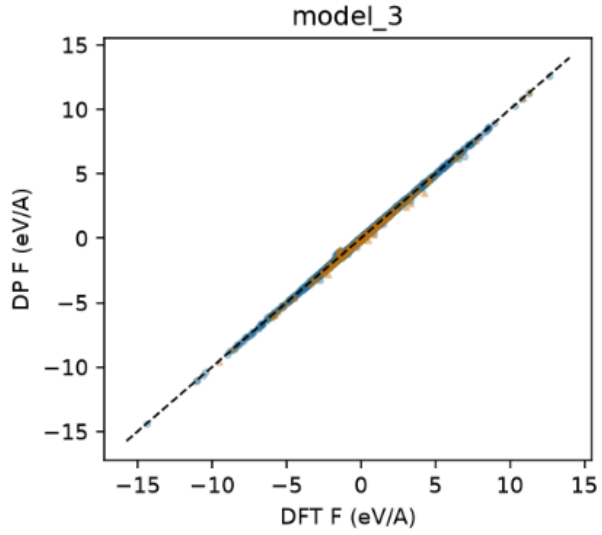

model_2
train
valid
DP E (eV)
DFT E (eV)
−1097
−1098
−1099
−1100
−1101
−1102
−1103
−1102
−1100
−1098

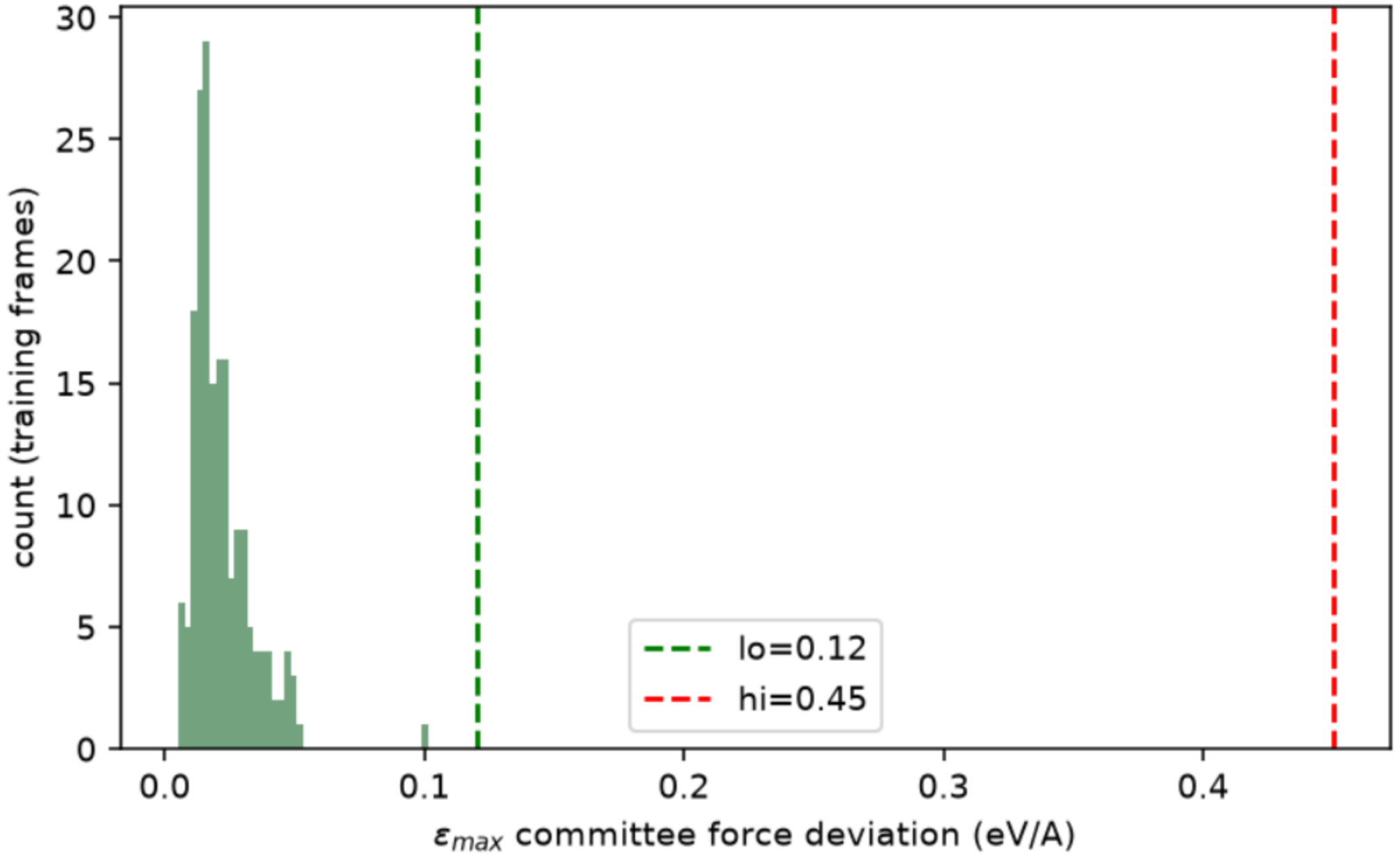

model_3
train
valid
DP E (eV)
DFT E (eV)
−1097
−1098
−1099
−1100
−1101
−1102
−1103
−1102
−1100
−1098


model_0
DP F (eV/A)
DFT F (eV/A)
15
10
5
0
−5
−10
−15
−15
−10
−5
0
5
10
15


model_1
DP F (eV/A)
DFT F (eV/A)
15
10
5
0
−5
−10
−15
−15
−10
−5
0
5
10
15


model_2
DP F (eV/A)
DFT F (eV/A)
15
10
5
0
−5
−10
−15
−15
−10
−5
0
5
10
15


model_3
DP F (eV/A)
DFT F (eV/A)
15
10
5
0
−5
−10
−15
−15
−10
−5
0
5
10
15


Committee uncertainty on the training set
count (training frames)
0
5
10
15
20
25
30
0.0
0.1
0.2
0.3
0.4
ε_{max} committee force deviation (eV/A)
lo=0.12
hi=0.45